

Muon's decay time and parity symmetry breaking



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1 Theoretical Introduction

1.1 Cosmic Rays and Atmospheric Muons

Primary Cosmic Rays are high energy astroparticles generated from outside of the Solar System and from Sun. When they enter the atmosphere, they interact with molecules in air, producing shower of secondary particles. Following interaction and decay processes, some particles reach the Earth surface.

Atmospheric muons are the most abundant charged particles reaching the surface. This is due to the small energy loss in the atmosphere, the small interaction cross section and a long lifetime, since its decay is mediated by weak interactions.

Atmospheric muons are mainly produced by the decay of secondary charged particles. The two main channels of production, with their branching ratios [1], are:

$$\begin{aligned}\pi^\pm &\rightarrow \mu^\pm \nu_\mu^{(-)} \quad B.R. : (99.98770 \pm 0.00004)\% \\ K^\pm &\rightarrow \mu^\pm \nu_\mu^{(-)} \quad B.R. : (63.56 \pm 0.11)\%\end{aligned}\tag{1.1}$$

1.2 Stopping power

When passing through matter, muons lose energy along their path inside the material. The average energy loss due to the interaction with matter, as a function of the energy of the muon, follows the expression:

$$\frac{dE}{dx} = a(E) + b(E) \times E\tag{1.2}$$

where $a(E)$ is the contribution due to ionization and atomic excitation of the material and $b(E) \times E$ is the contribution due to bremsstrahlung, pair production, photonuclear emission and radiative loss.

In Figure 1 are shown the stopping power curves by varying the energy of the muon, for 3 materials that will be used in this experiment. Polyvinyltoluene is the material of the detector used, while sodium chloride and aluminum will be used for specific measurements. Data are obtained from PDG [1].

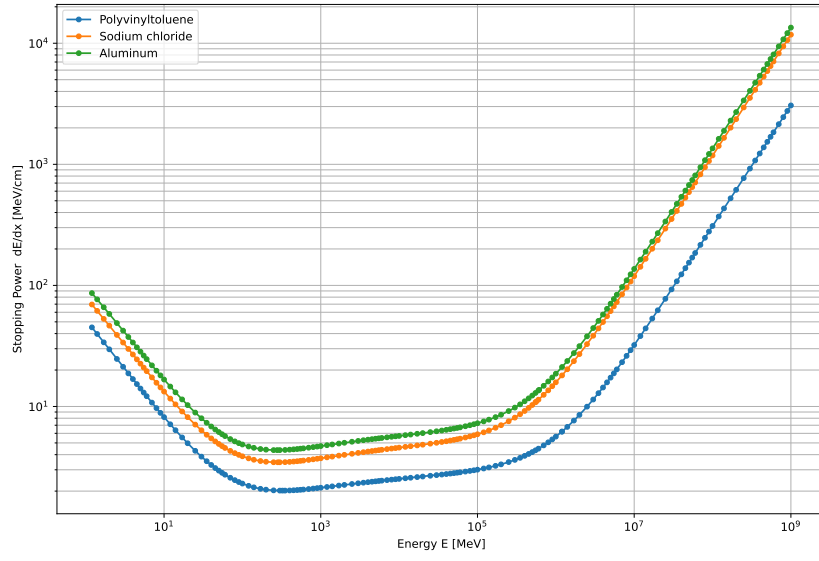


Figure 1: Stopping power, as a function of the muon energy, for different materials

By integrating the inverse of the stopping power we can find the maximum range of muons through every material, using Eq. (1.3). In Figure 2 is shown this relation as a function of energy.

$$\Delta x(E) = \int_0^x dx = \int_0^E \frac{dx}{dE} dE \quad (1.3)$$

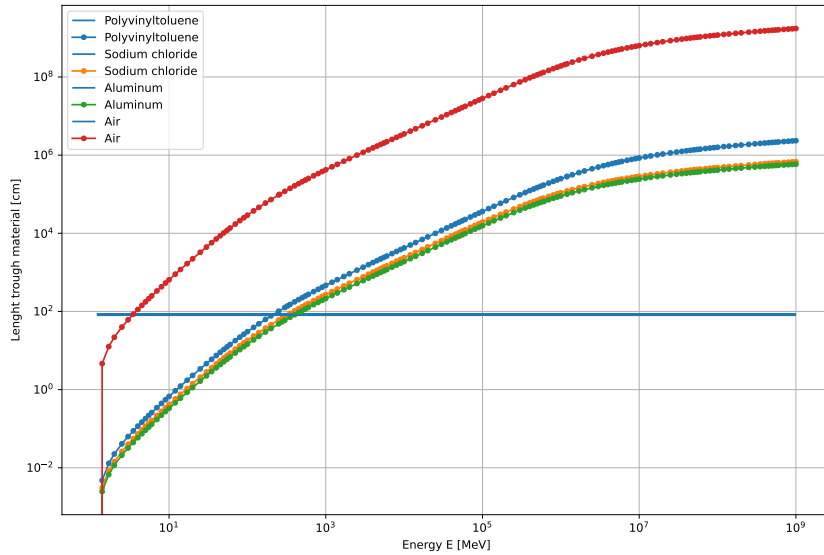


Figure 2: Length through matter for different material, and range in every material

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The horizontal line indicates the longest straight path a particle can perform in each material, taken as the diagonal of the block considered in our setup (for air 80 cm are taken for comparison **CONTROLLARE VALORI SCRITTI RISPETTO A QUELLI USATI NEI GRAFICI**).

From Figure 2 we can extrapolate the energy range of stopped muons in the materials shown, taking all the energies for which the maximum length is under the line drawn in the graph.

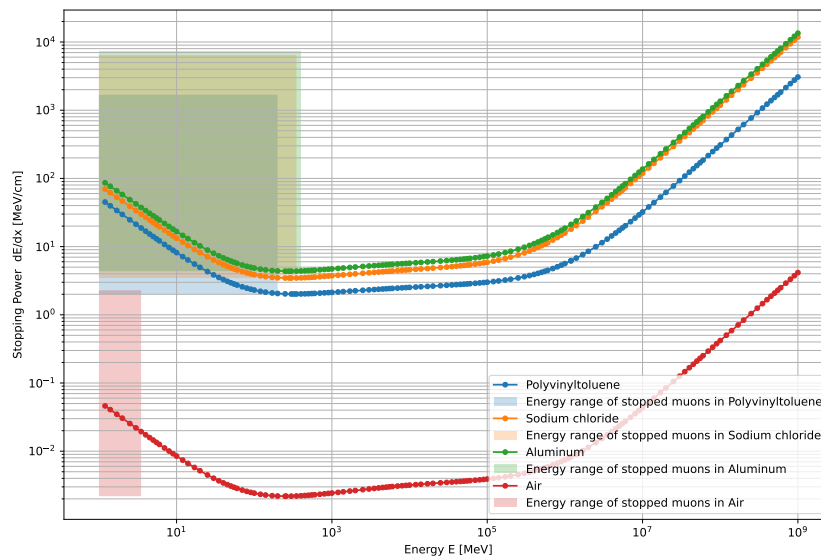
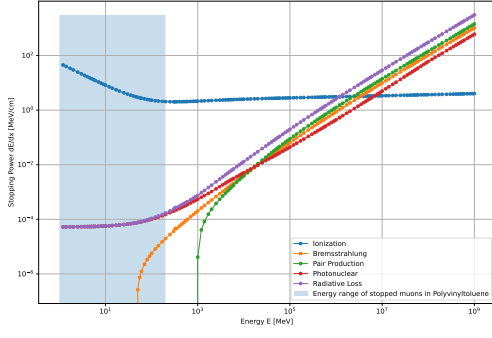


Figure 3: Stopping power and energy ranges, as a function of the muon energy, for different materials

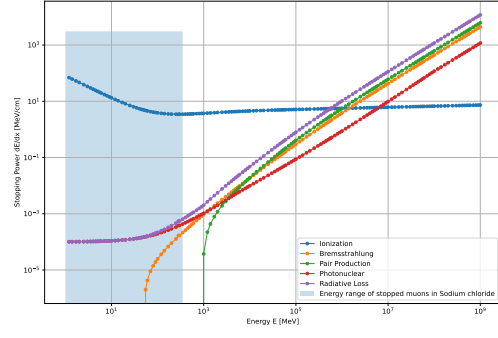
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In Figure 3 we can see the energy ranges thus obtained on the stopping power graph. This clearly shows that, for similar lengths, air adds a negligible contribute, since it has a smaller energies interval that can stop muons (with the same path length) and its stopping power is around 2 orders of magnitude lower than the other materials considered.

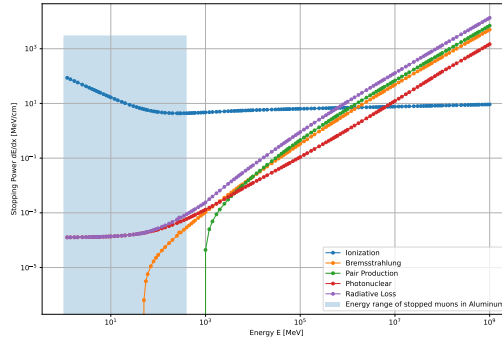
We can now use this estimated energy range to also see the different contributions of the stopping power we enlisted before. In Figure 4 are showed all the different contributions for the 3 materials we are interested in.



(a) Polyvinyltoluene



(b) Sodium chloride



(c) Aluminum

Figure 4: Stopping power contributions for different materials

We can notice how, at low energies, the contribution of ionization is at least 4 orders of magnitude more intense than the other contributions. Therefore we can assume that, for the energy range of our interest, only the ionization contribution can be considered.

1.3 Properties of Atmospheric Muons

Since the interaction with matter of μ^- and μ^+ is different, it is important to know the so called “muon charge ratio”, defined as the ratio of positive and negative atmospheric muons. Different measures showed a muon charge ratio varying with the energy (as shown in [2]). For low energy muons (≈ 1 GeV) the muon charge ratio $R_\mu = 1.18 \pm 0.02$ [3]. This gives the probabilities to have a μ^+ and μ^- :

$$P_+ = (54.1 \pm 0.4)\%$$

$$P_- = (45.9 \pm 0.4)\%$$

Atmospheric muons' angular distribution depends on their energy, as shown in Figure 5.

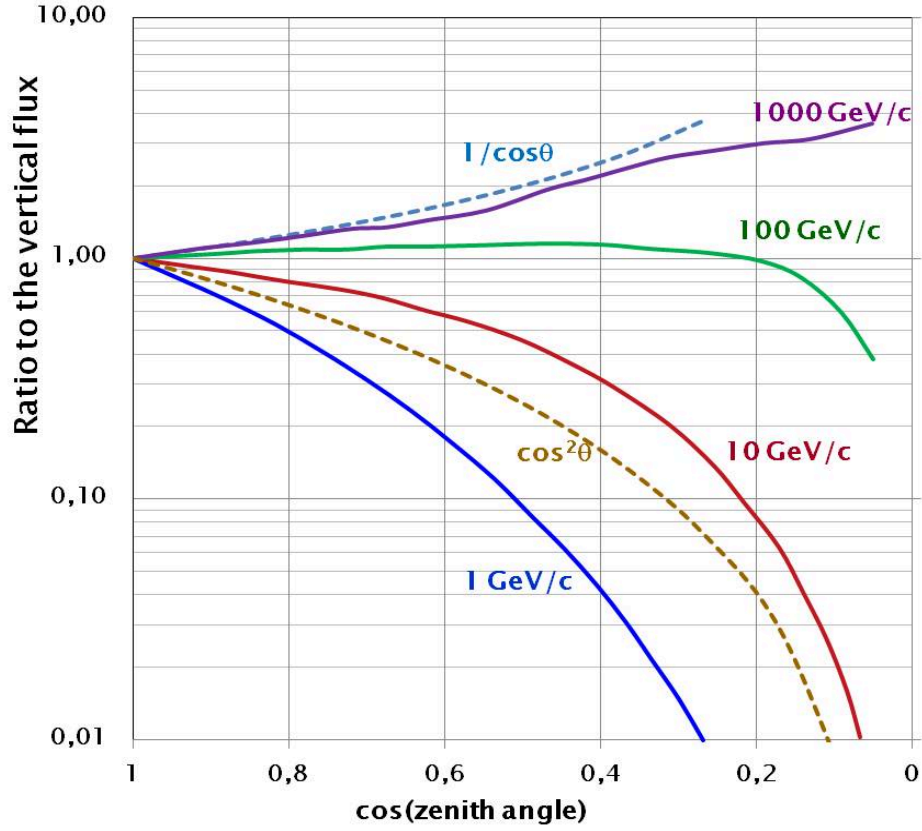


Figure 5: Angular distributions of atmospheric muon for different energies [4]

Muons with low energies (in which we are interested, as shown in the section Sec. 1.2) follow a $\cos^n \theta$ distribution, with $n \approx \frac{2}{3}$.

We are also interested in the muon flux at sea level for low energy particles. In Figure 6 is reported the graph of the integral (in $cm^{-2}sr^{-1}s^{-1}$) and differential (in $cm^{-2}sr^{-1}s^{-1}\left(\frac{GeV}{c}\right)^{-1}$) muon spectra, as reported by *Rastin* [5].

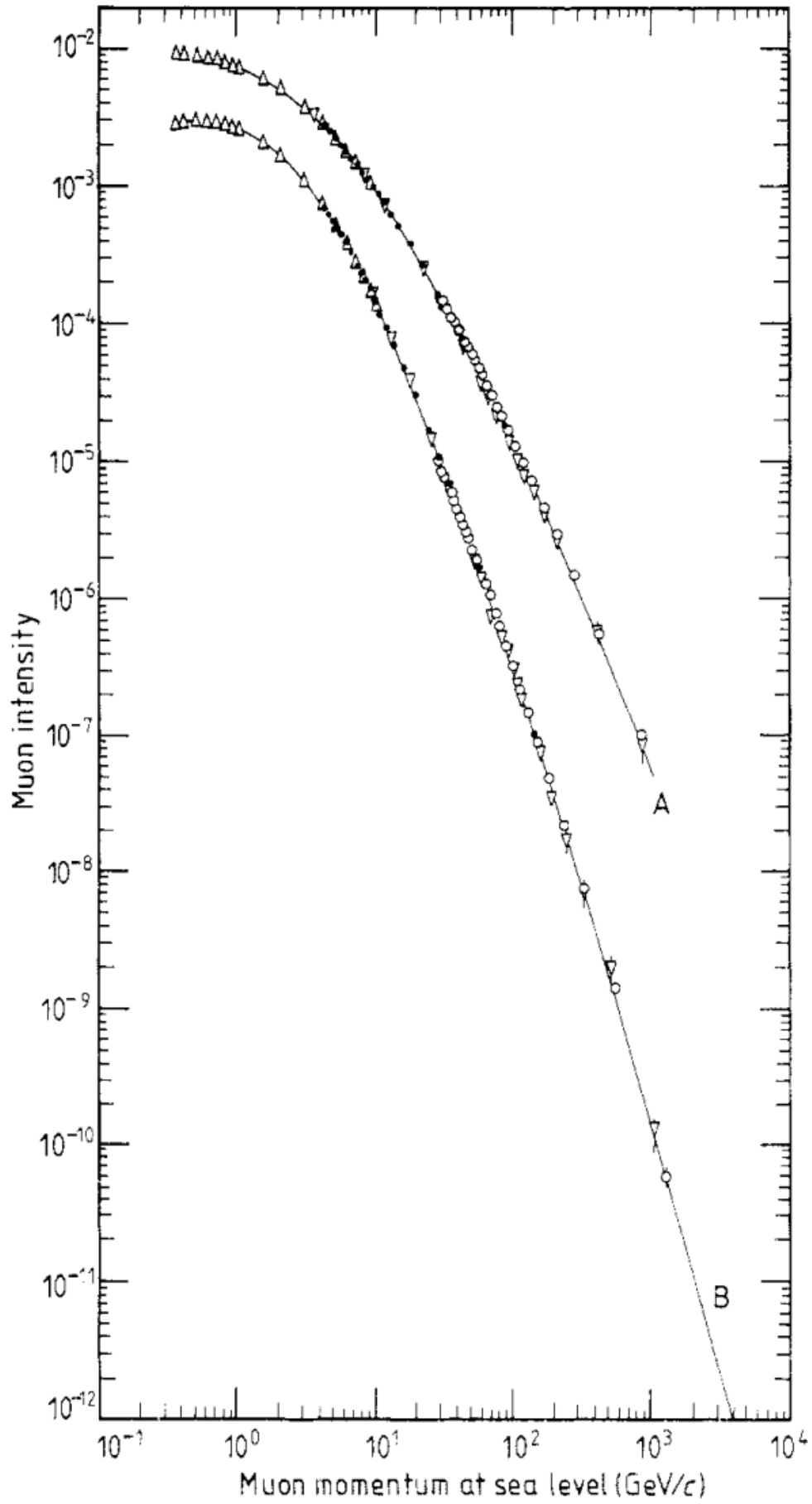


Figure 6: Integral (A) and differential (B) muon spectra for vertical direction [5]

SCRIVERE QUAL'È LA DIFFERENZA TRA DIFFERENTIAL E INTEGRAL, O È ABBASTANZA CHIARO? ANCHE PERCHÈ USIAMO INTEGRALE?

From the image we can observe that, for muons with momentum below 1 GeV, the flux looks almost constant. At exactly 1 GeV we have an integrated flux $7.29 \times 10^{-3} cm^{-2} sr^{-1} s^{-1}$.

1.4 Muons decay

Muon decays mainly ($\approx 100\%$) into an electron and two “neutrinos” ($\nu_\mu + \bar{\nu}_e$ for μ^- , and $\bar{\nu}_\mu + \nu_e$ for μ^+), as shown in Figure 7:

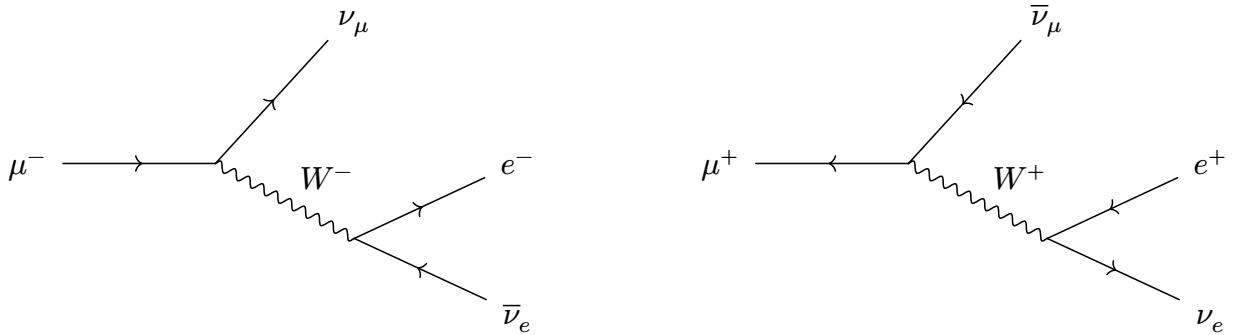


Figure 7: Feynman diagrams of muon's β decays

The other possible decays are higher order diagrams of this β decay, those of which imply a photon emission [1]. This means that, in order to measure muon lifetime, only a time measurement of the produced electron is needed.

Muon spontaneous decay time distribution follows an exponential, with a mean life $\tau = (2.1969811 \pm 0.0000022)\mu s$ (as listed in PDG [1]). Due to the exponential distribution, the time in which the μ is generated is not needed, since every time offset can be reabsorbed in the coefficient of the distribution. Hence, also atmospheric muons can be used to measure μ mean lifetime.

1.5 μ^- interaction with matter

When a muon stops in matter, it is subject to electromagnetic potential of the nuclei. In this case, before decaying, the muon can be captured by an atom. Since it's not

an electron, the captured muon is not subject to the Pauli exclusion principle, and it occupies the innermost orbital. Since the muon is more massive than the electron, his $1s$ orbital is less wide than the electron's one. Hence, the PDF of the muon has a non-negligible overlap with the protons in the nuclei, causing an interaction among them. The tree-level diagram of the interaction between the muon and the proton is showed in Figure 8.

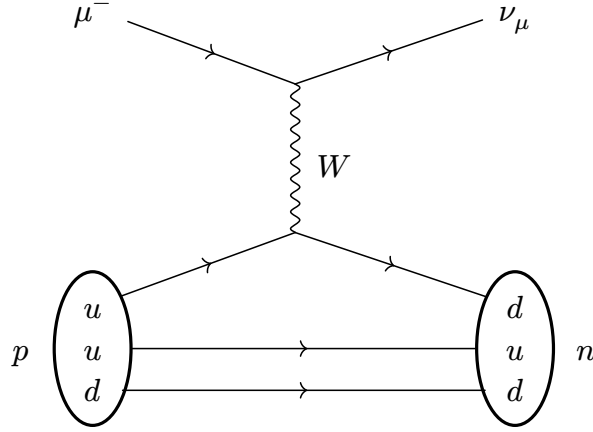


Figure 8: Feynman diagrams of μ^-p interaction

In this process, the muon rest energy gets redistributed among the neutrino and the neutron. Due to the neutrino is non-interacting, we will not be able to see it, so we have to search for some evidence of the excitation of the neutron, since it will gain some energy from the muon mass. The excited neutron goes back to its fundamental state through different paths: it can simply emit a photon or it could lose its energy through internal conversion, emitting one of the orbital electrons of the atom. As we will discuss later ([REF ALLA SEZIONE SE SERVE](#)) we will only be able to detect particles with a released energy (inside the detector) above a certain threshold. Since photons from radiative emission have energies comparable with ambient photons, that we already exclude because they form a disturbing background, we will not be able to detect them.

The capture rate of the muon will depend from the intensity of the EM interaction with the nuclei, so will depend on the Z_{eff} , as discussed by *Primakoff* [6]. The simplified Primakoff formula is reported in Eq. (1.4)

$$\Lambda_{\text{cap}}(A, Z) = Z_{\text{eff}}^4 X_1 \left[1 - X_2 \left[\frac{A - Z}{2A} \right] \right], \quad (1.4)$$

where X_1 is related the muon capture rate in hydrogen, X_2 is a factor that consider the Pauli exclusion principle in the nuclei, and Z_{eff} is obtained as discussed by *Ford and Willis* [7].

We are interested in the capture rate and in the mean life of captured muons inside Aluminum and Sodium Chloride. In advance, we should also consider the elements inside the scintillator as a possible nuclei capable of capturing the muon. But, as reported by *Suzuki et al.* [8], in both Hydrogen and Carbon μ^- has lifetimes similar to the vacuum one, with rates at least one order of magnitude smaller, that we are not able to distinguish. In Table 1 are reported those data from *Suzuki et al.*[8], and the Z_{eff} .

CARBONIO HA TAU=2.02, VS TAU_FREE=2.2. VA BENE COME ASSUNZIONE?

Element	Z_{eff}	$\tau(\text{ns})$	R (MHz)
Al	11.48	864 ± 2	0.662 ± 0.003
Na	9.95	1204 ± 2	0.3772 ± 0.0014
Cl	14.24	560.8 ± 2.0	1.333 ± 0.006

Table 1: Muon capture data for different elements, obtained from [8].

In order to obtain the decay time of captured muon in Sodium Chloride, we should take the weighted mean of the τ of Na and Cl, taking into account the relative capture rate of those two elements.

$$\tau_{\text{NaCl}} = \frac{\sum_{i=1}^2 R_i \tau_i}{\sum_{i=1}^2 R_i} \quad \sigma_{\tau_{\text{NaCl}}} = \sqrt{\sum_{i=1}^2 \left[\left(\frac{d\tau_{\text{NaCl}}}{dR_i} \sigma_{R_i} \right)^2 + \left(\frac{d\tau_{\text{NaCl}}}{d\tau_i} \sigma_{\tau_i} \right)^2 \right]}$$

with $i = 1, 2$ indicating, respectively, Na and Cl. By performing those calculations, we obtain

$$\tau_{\text{NaCl}} = (702.7 \pm 1.7) \text{ ns.} \quad (1.5)$$

Meanwhile, to obtain the total decay rate of captured muons in Sodium Chloride, we take the mean of the two rates.

$$R_{\text{NaCl}} = \frac{\sum_{i=1}^2 R_i}{2} \quad \sigma_{R_{\text{NaCl}}} = \sqrt{\sum_{i=1}^2 \left(\frac{dR_{\text{NaCl}}}{dR_i} \sigma_{R_i} \right)^2}$$

with $i = 1, 2$ indicating, respectively, Na and Cl. By performing those calculations, we obtain

$$R_{\text{NaCl}} = (0.855 \pm 0.003) \text{ MHz}. \quad (1.6)$$

Using those rates, we can now obtain the rate of muons, in each material, that decays freely or gets captured. The rate of the free decay is given by the τ of the muon.

$$R_{\text{free}} = \Gamma_{\text{tot}} = \frac{1}{\tau_{\mu}} \quad \sigma_{R_{\text{free}}} = \frac{\sigma_{\tau}}{\tau_{\mu}^2}$$

Using $\tau_{\mu} = (2.1969811 \pm 0.0000022)\mu s$ we obtain

$$R_{\text{free}} = (0.4551701 \pm 0.0000005) \text{ MHz}. \quad (1.7)$$

In a given material with rate of capture R_{mat} , we can compute the percentage of muons that decay freely and through capture. We only consider the error on R_{mat} since the error on R_{free} is various order of magnitude lower than the σ of the materials we are considering for this experiment.

$$R_{\text{free}}^{\%} = \frac{R_{\text{free}}}{R_{\text{free}} + R_{\text{mat}}}, \quad R_{\text{mat}}^{\%} = \frac{R_{\text{mat}}}{R_{\text{free}} + R_{\text{mat}}}, \quad \sigma_{R^{\%}} = \frac{R_{\text{free}} \sigma_{R_{\text{mat}}}}{(R_{\text{free}} + R_{\text{mat}})^2}.$$

Material	$R_{\text{free}}^{\%}$	$R_{\text{mat}}^{\%}$	$\sigma_{R^{\%}}$
Al	40.7%	59.3%	0.1%
NaCl	34.74%	65.26%	0.08%

Table 2: Probability that a muon decays freely or through capture in different materials.

1.6 μ^+ interaction with matter

When a positive muon gets thermal inside a material, it can capture an electron to form the muonium $M = \mu^+ e^-$. Due to the instability of μ^+ the muonium decays as shown in Figure 9.

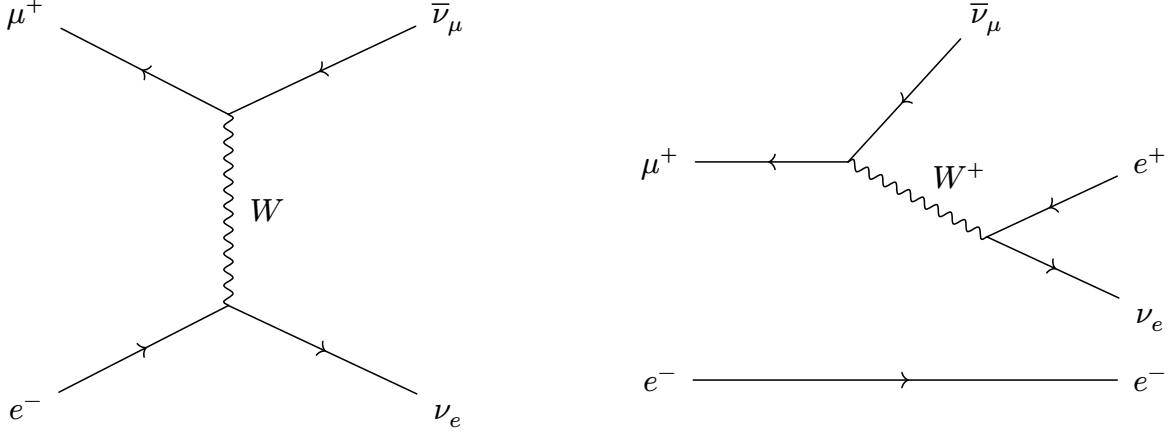


Figure 9: Feynman diagrams of muonium decays

Although the invisible decay still hasn't been observed, Standard Model predictions suggest the process would have a branching ratio $\text{Br}(M \rightarrow \nu_e \bar{\nu}_\mu) = 6.6 \times 10^{-12}$ [9]. The muonium preferably ($\approx 100\%$) decays into $e^+ e^- \nu_e \bar{\nu}_\mu$. The decay time is similar to the free anti-muon decay, but gets some corrections due to the interaction of the two particles. As discussed by *Czarnecki et al.* [10], the leading effects, caused by the motion of the μ^+ in the rest frame of the muonium, generate a correction term

$$\tau_M = \tau_\mu \left(1 + \frac{\alpha^2 m_e^2}{2m_\mu^2} \right), \quad \left(\frac{\alpha^2 m_e^2}{2m_\mu^2} \right) \approx 6 \times 10^{-10}.$$

Since the correction is really small, we will not consider it, and simply take $\tau_M = \tau_\mu$.

1.7 Up-Down Asymmetry

Atmospheric muons are mainly produced from a two body decay of charged kaons and pions, as already mentioned in Sec. 1.1. Both those mesons are spinless particles, so the final state has to be a singlet state of spin. In the meson's rest frame, this means that the muon and the neutrino must have same helicity (opposite spin direction). In SM, neutrino are massless, so ν must be left-handed and $\bar{\nu}$ must be right-handed. Therefore μ^- must be produced right-handed, while μ^+ must be produced left-handed. But, since π and k are highly energetic, the products of the decay will receive a Lorentz boost, and helicity is not Lorentz invariant. This means that is possible to see at sea level some muons with the “wrong” polarization.

Due to the V-A structure of the weak coupling, there is a correlation between the direction of the produced $e^\pm$ and the spin orientation of the $\mu^\pm$, with a differential decay width:

$$\frac{d^2\Gamma_\pm}{dx d\Omega} = \frac{G_F^2 m_\mu^5}{192\pi^3} x^2 \left[(3 - 2x) \pm P_\mu (2x - 1) \cos \theta \right] \quad (1.8)$$

with $G_F \approx (1.1663787 \pm 0.0000006) \times 10^{-5} \text{GeV}^{-2}$ Fermi coupling, $m_\mu \approx (105.6583755 \pm 0.0000023) \text{MeV}$ mass of the muon, $x = \frac{2E_e}{m_\mu}$ energy fraction of the $e^\pm$, θ angle between $\mu^\pm$ spin direction and direction of the $e^\pm$ momentum and P_μ polarization of the $\mu^\pm$, that, for an energy range of cosmic muons $E = (0.75 - 1.17) \text{GeV}$, is $P_\mu = -0.33 \pm 0.04$ [11]. By integrating over $x = [0, 1]$ to consider all possible energy fractions, we obtain: **METTI CSI = 1**

$$\frac{d\Gamma_\pm}{d\cos\theta} = \frac{\Gamma_\mu}{2} [1 \pm A' \cos \theta] \quad (1.9)$$

with $A' = \frac{1}{3} P_\mu \xi$ integral asymmetry [12], and ξ is one of the Michel parameters, that describe the phase space distribution of charged leptons decays into lighter leptons. Now we can integrate over θ , in order to separate the upward decays and the downward ones. Since, in the lab-frame, μ^- are mainly right-handed, the electron goes upward for $\theta = [\pi, \frac{\pi}{2}]$. Meanwhile, since μ^+ are mainly left-handed, the positron goes upward for $\theta = [0, \frac{\pi}{2}]$. This means that upward $U(t)$ and downward $D(t)$ decays will follow these relations

$$U(t) = U_0 \left(1 + \frac{A'}{2} \right) e^{-\frac{t}{\tau_\mu}} \quad (1.10)$$

$$D(t) = D_0 \left(1 - \frac{A'}{2} \right) e^{-\frac{t}{\tau_\mu}} \quad (1.11)$$

obtained assuming no depolarization effects or background events.

Now we can define the up-down asymmetry as

$$A = \frac{U(t) - D(t)}{U(t) + D(t)} = \frac{(U_0 - D_0) + \frac{A'}{2}(U_0 + D_0)}{(U_0 + D_0) + \frac{A'}{2}(U_0 - D_0)} \quad (1.12)$$

If $U_0 = D_0$, we would obtain $A = \frac{A'}{2}$.

In order to measure this asymmetry, one could put a magnetic field B with a component orthogonal to the spin direction. By doing so, the spin will precess around the

magnetic field axis, changing its direction. This process follows the Larmor precession, which tells us the spin will precess around the B direction with a frequency:

$$\omega_\mu = -\frac{egB}{2m_\mu} \quad (1.13)$$

with e electric charge and $g \approx 2$ gyromagnetic factor (neglecting corrections above tree-level). This effect makes the polarization time dependent with a periodic law. Consequently $A'(t) = \frac{1}{3}P_\mu \xi \cos(\omega_\mu t)$.

**TODO: interpolazione con funzione ref
polarization_of_cosmic-ray_muons? (con ag-
giunta di termine di depolarizzazione $\sin(ft)/ft$)**

2 Instrumentation and Modular Electronics

2.1 Experimental Methodology and Workflow Architecture

In order to transform raw physical interactions into reliable datasets we used the so-called “modular” approach: all acquisition process is mediated by many different components and each of them has its own porpoise **PURPOSE?**. So, rather than treating an acquisition setup as a rigid single instrument, we built our setup upon standard modular electronics that we arranged together.

This “modular” approach requires 4 steps to be able to collect reliable data:

1. **Design and Functional Assembly:** Conceptualizing the signal path logic and physically staging the individual hardware modules within the crate. This is influenced by the available and functioning modules we had access to and our choices of what was more important for us (as will be discussed in Sec. 3).
2. **Component Characterization:** Separately testing and validating each hardware component to check if each of them works as we would expect.
3. **System Calibration and Integration:** Combining the different modules into an interconnected, synchronized chain. Tuning thresholds, time delays, cables, exc. so that an initial signal can go through all the chain and give a resulting signal. This step also verify what in step 1. was theorized.
4. **Data Acquisition:** Launching the data collection and digitization for offline physical analysis.

This approach is very efficient in pursuing troubleshooting for each module that can lead to instrument dependencies and electronic anomalies, which can be systematically identify and correct.

2.2 Experimental Infrastructure and Auxiliary Equipment

The experimental setup follows the **Nuclear Instrumentation Modules (NIM)** standard, using a dedicated crate to house and power the modular electronics. This crate provides the essential high-stability power rails (± 6 V, ± 12 V, ± 24 V) and ensures a common grounding scheme. This configuration allows the minimization of the electronic noise by preventing ground loops.

The following precision laboratory instruments have been used to test all the electronic modules to obtain a stable and reliable acquisition: **AND THEREFORE WILL NOT BE TESTED?**

- **Tabor 8020 Pulse Generator (20 MHz)** served as the primary signal source for module characterization, providing reproducible pulses with adjustable parameters to simulate real detector events.
- **Tektronix TDS 2014B Oscilloscope (100 MHz, 1 GS/s)**
- **Multimeter Fluke 115 True RMS**

2.3 Data Acquisition System and NIM Modules

The signal processing chain is designed to transform analog pulses into calibrated logic signals for real-time coincidence in order then to be processed and digitalized. The workflow is built by a suite of NIM-standard modules, which handle everything from signal replication to temporal logic. The instrumentation used is organized as follows:

- **Discriminator Mod. N 96 (CAEN):** Converts negative analog signals into standardized NIM logic pulses based on a tunable voltage threshold, serving as the primary trigger source.
- **Logic Fan-In/Fan-Out Mod. 429A (LeCroy):** Provides signal duplication and logical AND/OR summation. **AND/OR OR ONLY OR? CONTROLLARE**
- **Octal Fan-Out Mod. 430 (LeCroy):** An 8-channel distribution module used to replicate logic signals across various branches of the acquisition chain.
- **Logic Unit Mod. N405 (CAEN):** A programmable 4-channel unit used to perform Boolean operations (AND, OR, XOR).
- **Dual Timer N93B (CAEN) & Dual Delay:** This module provides precise control over the logic pulse duration and temporal alignment, allowing for the synchronization of signals from different detectors. **SONO 2 MODULI**
- **Counter Mod. N1145 (CAEN):** Used for real-time monitoring of event rates and total counts, providing an immediate visual check of the experimental flux. Mostly used during the “testing” part of the experiment.
- **Digitizer DT5720B (CAEN):** A 12-bit ADC with a 250 MS/s sampling rate, performing digital charge integration and waveform reconstruction for offline analysis.
- **Power Systems:** A Programmable Power Supply and a High Voltage (HV) Power Supply providing the stable bias voltages required for the detectors and the high-stability current for the NIM crate rails.

2.4 Logic Verification and Signal Distribution

The acquisition chain was initially tested to ensure the correct distribution of signals and proper synchronization across all timing branches. During this phase we proceeded in testing and characterizing each of the modules in order to ensure the proper functioning and performance. The first ones we tested were the **LeCroy 429A Fan-In/Fan-Out** and **LeCroy 430 Octal Fan-Out** modules: we fed the logic NIM signals into the modules to characterized them and to verify their preservation of fast rise times (< 2 ns). Testing confirmed that output amplitudes remained stable across multiple $50\ \Omega$ loads, allowing for the simultaneous distribution of signals to the trigger logic, the

monitoring counters, and the digitizer inputs. During this stage, signal “ringing” was successfully mitigated by terminating all oscilloscope channels with $50\,\Omega$ DC coupling and capping unused inputs with LEMO terminators to prevent inductive crosstalk.

To test the counter **CAEN Mod. N1145** we used the following technique: knowing the frequency of a periodic signal generated by the pulse generator and setting a fixed acquisition time it was possible to check the reliability of the counts displayed by the counter. **SU DRIVE ABBIAMO LE FOTO DELLA CALIBRAZIONE**

2.5 Characterization of the Discriminator

Mod. N96

Prior to the selection of the current module, the **CAEN Mod. N417** was initially considered; however, resulted unsuitable for the experimental requirements due to significant hardware degradation and stability issues. In trying to test the module several channels resulted damaged, including compromised adjustment screws that hindered precise calibration, and the threshold levels proved to be highly unstable over time. In addition, the N417 module failed to meet the dynamic range requirements of the experiment, as its maximum threshold was insufficient compared to the 250 mV signals targeted for rejection. Therefore, the **CAEN N96** was adopted to provide the necessary mechanical reliability and threshold stability.

A fundamental stage of the characterization for this module involved the linearity calibration of its central channels (CH4 to CH7, the ones we used). In fact, although all the channels have been tested, only the channels 4 through 7 were reliable because the other channels we observed several problems, such as compromised adjustments screws and stability issues. This process was essential to establish the exact relationship between the manual potentiometer settings (V_{th}) and the actual minimum input amplitude (V_{in}) required to fire the discriminator. The experimental setup used the **Tabor 8020** to generate negative square pulses, which were split via a Fan-Out: one branch was sent to the oscilloscope for amplitude monitoring, while the other was fed into the **N96** discriminator. Thresholds were swept from -50 mV to -225 mV and measured with a precision multimeter. To account for mechanical hysteresis and ensure consistency, a

“unidirectional approach” was adopted: we set each target threshold reaching it from a lower value to an higher one.

The resulting data were modeled using the following linear regression:

$$V_{in} = a \cdot V_{th} + b \quad (2.1)$$

TODO: scrivere come stimati gli errori

In order to estimate the errors, we measured the oscillation of the set threshold. We measured an oscillation of $\Delta V_{th} \approx 3$ mV **CHECK**, therefore we have assumed an uniform distribution for the error with a $\sigma = 3$ mV and decided to apply to all measurements.

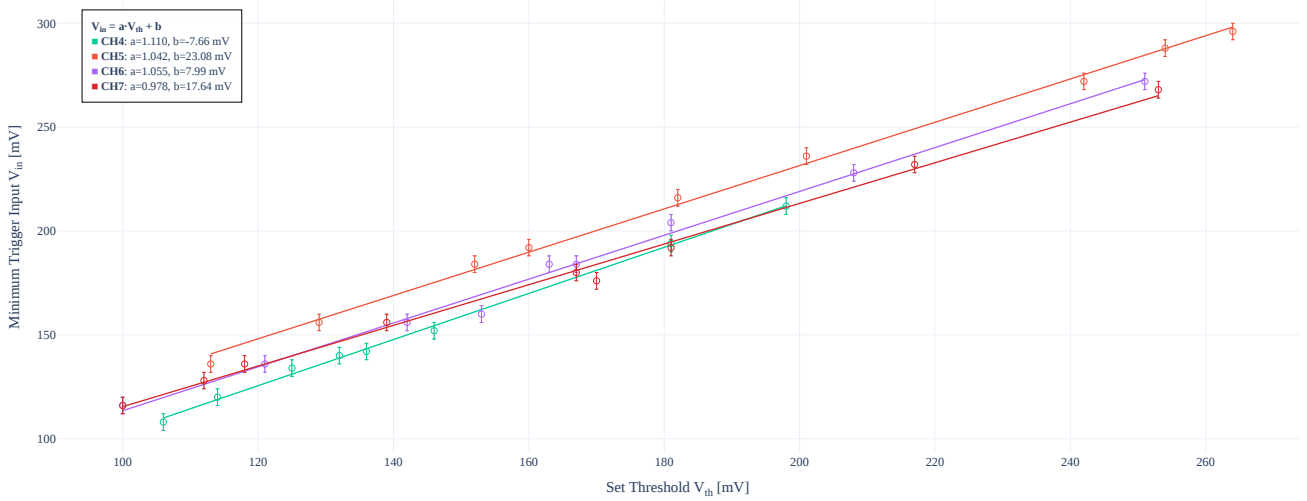


Figure 10: Linear regression of the N96 Discriminator threshold calibration for channels CH4-CH7.

To evaluate the goodness-of-fit and the consistency of the discriminator response, a statistical analysis based on the reduced Chi-squared ($\tilde{\chi}_\nu^2$) was performed. The results, summarized in Table 3, confirm a good linearity, as the $\tilde{\chi}_\nu^2$ values remain close to unity.

UN χ_ν^2 DI 3.82 NON È ESATTAMENTE CLOSE TO UNITY

Channel	a (Slope)	σ_a	b (Intercept)	σ_b	$\tilde{\chi}_\nu^2$
CH4	1.110	0.035	-7.66	5.04	0.68
CH5	1.042	0.032	23.08	6.13	0.95
CH6	1.055	0.053	7.99	8.98	1.70
CH7	0.978	0.039	17.64	6.64	0.89

Note: All voltage measurements (b, σ_b) are expressed in mV. Input voltage uncertainty $\sigma_{V_{in}} = 3$ mV.

Table 3: Fit parameters, uncertainties, and reduced Chi-squared values for the N96 discriminator channels.

Channel	a (Slope)	σ_a	b (Intercept)	σ_b	$\tilde{\chi}^2_\nu$
CH4	1.110	0.023	-7.66	3.36	1.53
CH5	1.042	0.021	23.08	4.09	2.14
CH6	1.055	0.053	7.99	8.98	1.70
CH7	0.978	0.026	17.64	4.43	2.01
<i>Note: All voltage measurements (b, σ_b) are expressed in mV. Input voltage uncertainty $\sigma_{V_{in}} = 2$ mV for CH4, CH5, CH7 and $\sigma_{V_{in}} = 3$ mV for CH6.</i>					

Table 4: Fit parameters, uncertainties, and reduced Chi-squared values for the N96 discriminator channels.

The analysis revealed slopes (a) near unity, with slight deviations (0.978 to 1.110) reflecting internal gain factors specific to each channel. These values, alongside the intercepts (b), serve as calibration constants to accurately map physical thresholds during data acquisition. Notably, CH4 exhibited non-linear saturation for thresholds exceeding 250 mV, causing the operational range to be strictly limited. Therefore, since only 3 channels were needed, we used CH5, CH6 and CH7.

2.6 Characterization of the Dual Timer Mod. N93B

To verify the gate-generation capabilities and the temporal precision of the CAEN N93B Dual Timer, an overlap analysis was performed using two decoupled channels monitored via the oscilloscope. A master logic pulse from the generator was split: the first branch was routed directly to Channel 1 (CH1) to act as the reference trigger, while the second branch was used to trigger the N93B Dual Timer, with its output monitored on Channel 2 (CH2).

As illustrated in the captured waveform, **AGGIUNGERE FOTO + REF** both channels exhibit stable NIM logic levels transitioning from a baseline of 0 V down to approximately -0.85 V. The dual timer introduced a clean, programmable propagation delay of exactly 18 ns relative to the reference signal's falling edge at $t = 0$ ns. **NOT EXACTLY, 18+-INCERTEZZA** The testing of this module is important in order to maintain a good defined gate window in order to have reliability in the chain acquisition.

To map the response of the timer channels, a master trigger pulse was fed into the input, and the resulting output logic pulse durations were systematically measured across all coarse range selector positions using the oscilloscope. The calibration results for the four output channels (CH1–CH4) are summarized in Table 5.

Nominal Set- ting	Timer 1 (CH1)	Timer 1 (CH2)	Timer 2 (CH3)	Timer 2 (CH4)
50 ns	50 ns	50 ns	70 ns	3 ns
200 ns	138 ns	148 ns	192 ns	50 ns
1 μ s	720 ns	800 ns	1.9 μ s	60 ns
10 μ s	6.4 μ s	6.6 μ s	20 μ s	50 ns
100 μ s	24 μ s	30 μ s	163 μ s	50 ns
1 ms	56 μ s	76 μ s	1.5 ms	50 ns
10 ms	620 μ s	1.4 ms	15 ms	50 ns
100 ms	7 ms	8 ms	175 ms	50 ns
1 s	49 ms	127 ms	2 s	50 ns
10 s	—	—	—	50 ns
<i>Note: Measured output pulse widths correspond to discrete coarse knob selector positions on the CAEN N93B module.</i>				

Table 5: Measured output pulse widths for each coarse knob position on the CAEN Dual Timer Mod. N93B across all four output channels (CH1–CH4).

As detailed in Table 5, channels CH1, CH2, and CH3 scale monotonically with the rotary selector position, providing broad gate-width coverage from 50 ns up to full seconds. In contrast, Channel 4 (CH4) remains fixed at a narrow, fast pulse width (50 ns) INTENDI ~ ? across higher switch positions, effectively operating as an auxiliary prompt logic output.

2.7 Characterization of the Logic Unit Mod.

N405

This module has been tested in order to verify that the minimum temporal synchronization window for a logic operation such as AND or OR had a fixed value to guarantee overlapping of signals. In this prospective a timing overlap test has been conducted.

The procedure involved generating identical negative NIM pulses, which were split and routed through physical passive delay stages to introduce a controlled relative time shift. These time-stamped logic pulses were then fed into the input channels of the N405 unit. By progressively adjusting the delay lines, the overlap width between the signals was incrementally varied while the logic unit’s output was continuously monitored.

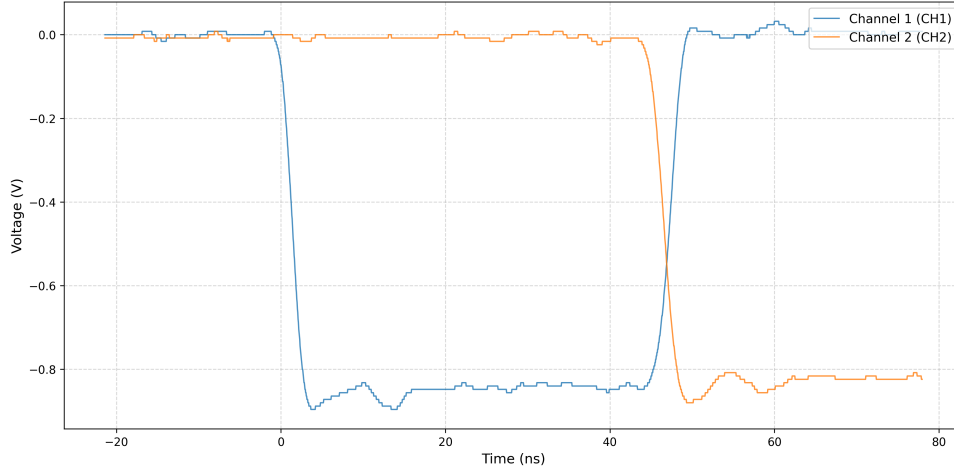


Figure 11: Oscilloscope screenshot displaying the coincidence window validation on the CAEN N405 Logic Unit.

Using the oscilloscope’s temporal cursors, the overlap between the falling edge of the trailing signal and the rising edge of the leading signal was measured to be exactly $\Delta t = 6$ ns. The test has been conducted in several stages and all confirmed that the Mod. N405 stably registers this 6 ns overlap without generating transient logic state anomalies or spurious “glitch” pulses. **MA I 6 NS È IL MINIMO PER FARE OPERAZIONI, NO? ANCHE MAGGIORE DI 6 VA BENE** This shape of the sharp edge temporal waveform in a very good index that the logic unit is capable of executing high-rate Boolean operations without drift, fulfilling a critical requirement for suppressing random background coincidences during the final data acquisition run.

2.8 Characterization of the Dual Delay Module

SEGNALE SDOPPIATO, VERIFICATO SULL OSCILLOSCOPIO CHE L IMPOSTATO FOSSE UGUALE A QUELLO VISUALIZZATO To verify the precise passive phase-shifting capabilities of the Dual Delay module, a time-delay

characterization was executed using the oscilloscope. The primary aim of this test was to accurately quantify the hardware-induced temporal shift and evaluate edge degradation or amplitude attenuation across the delay line.

The characterization was performed by feeding a master logic pulse into a Fan-In/Fan-Out module to duplicate the signal into two identical branches: a reference branch routed directly to Channel 1 of the oscilloscope, and a second branch passed through the passive Dual Delay module under test before being monitored on Channel 4.

Direct observation on the oscilloscope confirmed that the set delay value matched the measured time shift precisely. Furthermore, a side-by-side comparison of the waveforms verified that the input and output signals were virtually identical in profile; the delayed pulse preserved its sharp, sub-nanosecond rise time and full NIM voltage amplitude without suffering from edge slurring, distortion, or attenuation. This transparent propagation confirms the module's reliability for fine-tuning signal alignment across independent detection channels prior to coincidence logic processing. **ABBIAMO**

FOTO? ABBIAMO FOTO SOLO IN JPG NON IN CSV

2.9 Data Acquisition and Waveform Digitization

The final stage of the acquisition chain is the **CAEN DT5720B Digitizer**, a 12-bit ADC operating at a 250 MS/s sampling rate. The primary objective of the digitizer calibration was to establish a precise mapping between raw ADC counts and physical voltage levels (V) **V?**. Given the 12-bit resolution distributed over a 2 Vpp dynamic range, the theoretical conversion factor is determined by:

$$\Delta V = \frac{2V}{2^{12}} \approx 0.488 \text{ mV/count} \quad (2.2)$$

Data acquisition and visualization parameters were managed via the CAENScope software interface, where the acquisition window was configured with a record length of 2048 sampling bins per event. At a clock speed of 250 MS/s (corresponding to a 4 ns time step per bin), this arrangement yielded a comprehensive observation window of

8.192 μs , providing more than sufficient temporal margin to capture the entire evolution of the pulse alongside its surrounding baseline.

Before starting with the data collection, the software has been used and tested in different configurations in order to understand its behavior in terms of noise and offsets.

3 Electrical setup

The setup we chose to minimise dead time and pileup probability is the following:

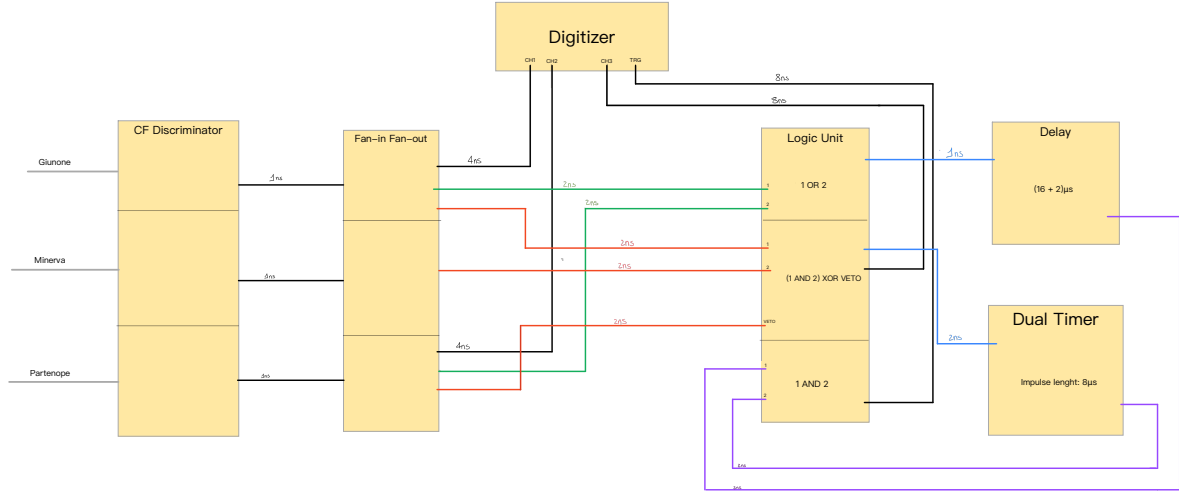


Figure 12: Muon flux as a function of energy of the muon

After studying the signal of all three PMT, we opted for a configuration in which Minerva (thin scintillator) is placed in the middle of the other two. **REF TO PMT**

SIGNALS STUDY This is due to the PMT signal showing a non ideal behaviour.

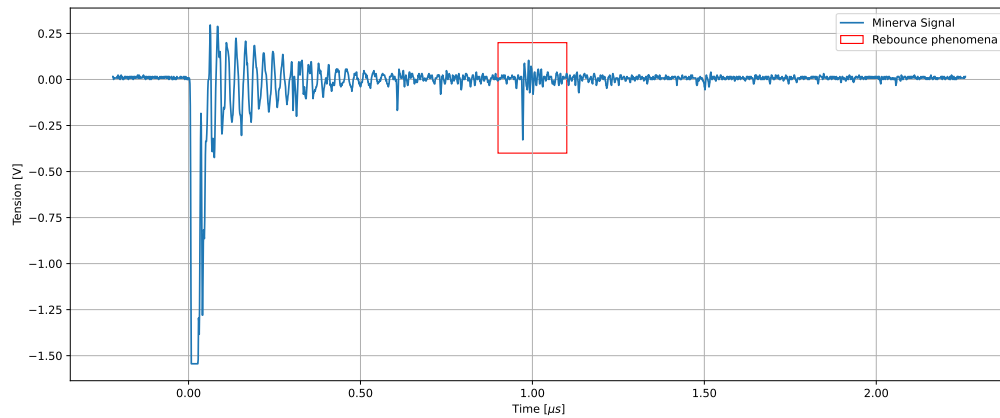


Figure 13: Example of the rebound phenomena in Minerva

QUESTA PARTE CHI LA FA? By placing Minerva in between the two others, we willingly choose to have a thinner layer of scintillator material in which the muon can decay, with the pro of having a lower number of spurious coincidence and therefore a higher S/N ratio.

The logic of the setup is done using a logic unit. It's used in three different parts of the setup:

- To detect the muon that stops in the middle scintillator
- To detect a particle that reaches one of the other two detectors
- To combine this two information in such a way to detect the electron produced after the decay of the muon

While the first two operation can be done with the sole usage of the logic unit, to realise the third operation we opted for the addition of a dual timer, thanks to which we are able to have more freedom of selecting a temporal range adequate to our measurement.

Another peculiarity of this setup is the connection with the digitizer: instead of using the muon signal as a trigger and then wait for the electron signal, we opted for the inverse setup, in which we use the electron signal as a trigger and then we use the cached data of the digitizer to look for the muon signal.

This last one does not improve the measurement or the errors but it reduces the probability of having an empty trace to almost 0. Which means that the size of the dataset is smaller and therefore easier to manage.

3.1 Sensibility

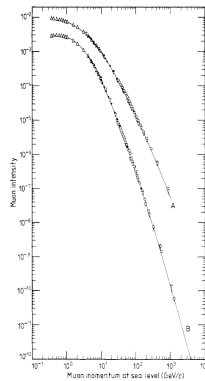


Figure 14: Muon flux as a function of energy of the muon

As we can see in figure Figure 14, the flux of muons for energies lower than 1GeV is about constant. Therefore it is easy to calculate the expected rate of muons given the energy range calculated previously **REFERENCE CALC RANGE ENERGIA**. By doing so, we expect a flux of muons

3.2 Time resolution and dead time

We chose to measure the time resolution of the setup by simulating an event with a wave generator. Giving to the discriminator a strictly positive square wave with a width comparable to the signal observed and a low frequency to avoid pileup effect, we were able to delay the signal of the bottom scintillator to simulate both the muon decay and the electron release with only one impulse from the wave generator.

It is enough to simulate the electron energy release only in the bottom scintillator since the electronic chain is completely symmetrical.

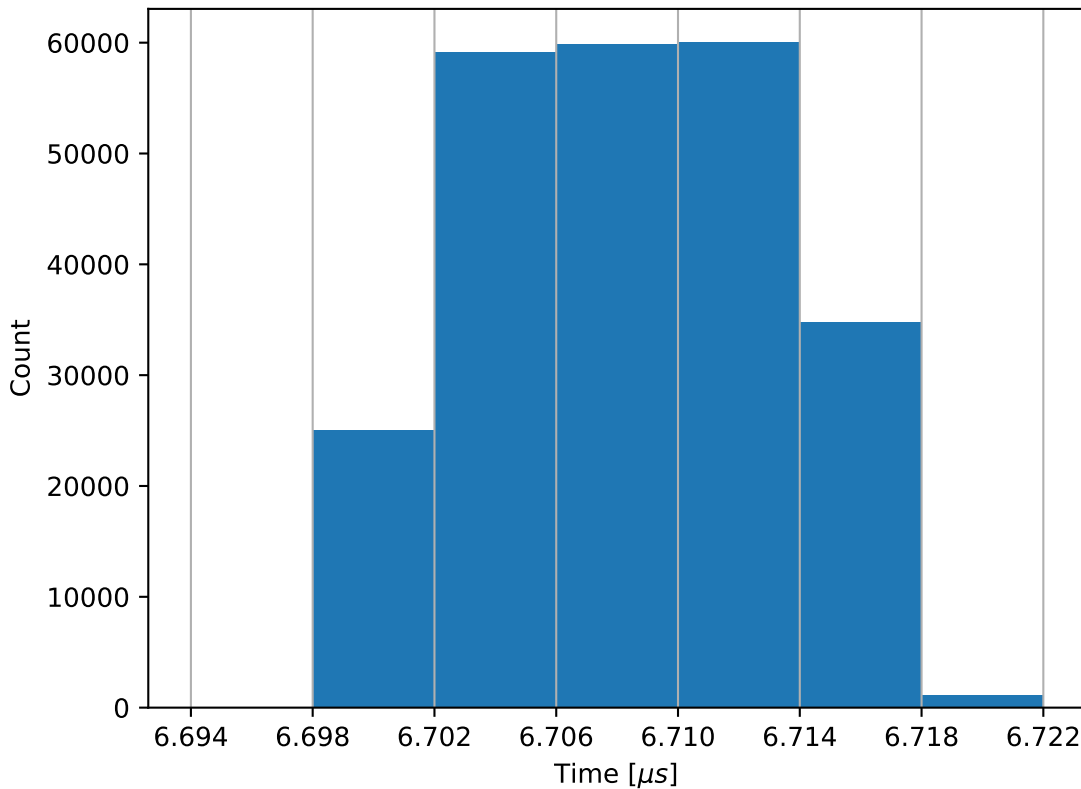


Figure 15: Measured time resolution of the whole setup

The result is a distribution about uniform of width $12ns \pm 4ns$, the uncertainty is fully determined by the frequency of the digitizer. This tells us the minimum error that we can expect from our setup without considering the scintillators.

Another statistically relevant characteristics of the setup is the dead time. Since the setup used previously was not useful for an accurate measurement of the dead time, we opted for calculating this factor using actual data. This is easily done using actual

data: if we express the impulse given by the muon stopping in the middle scintillator and we look for the closest electron signal, what we find is the exact measurement of the dead time of the setup

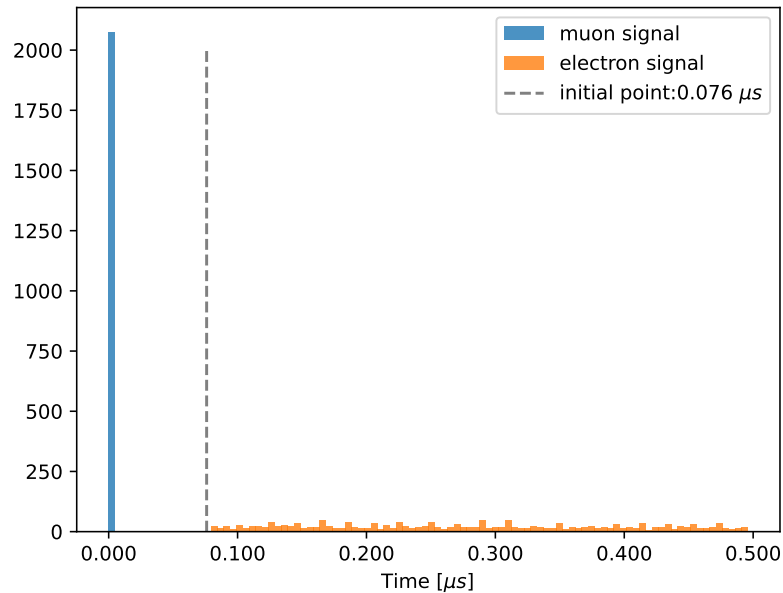


Figure 16: zoom on an initial part of a dataset

We can easily measure it given a dataset and it turns out to be: $76ns \pm 4ns$ where the uncertainty of this value is again dominated by the frequency of the digitizer.

3.3 Pileup probability

IO METTEREI LA PARTE DI STIMA DEL FLUSSO QUI (O SE PREFERITE NELLA PARTE DI INTRODUZIONE TEORICA) E TOGLIEREI LA SEZIONE 3.1

Given the expected flux used in the previous section, the dead time calculated previously and the fact that the apparatus is non paralyzable to calculate the probability of pileup.

We can clearly see that the percentage of pileup is very close to 0 and therefore can be neglected.

4 Scintillators

4.1 Technical characteristics

This chapter details the physical and operational properties of the plastic scintillators used. These detectors, based on a Polyvinyltoluene (PVT) polymer base, are known for their high speed, excellent light yield, and versatility in high-energy physics applications.

The scintillators used (specifically the EJ-200 / BC-408 and EJ-204 / BC-404 series) are organic plastic scintillators. Their operation is based on the excitation of the polymer base by ionizing radiation, followed by the emission of fluorescence photons. Polyvinyltoluene provide high light conversion efficiency. The primary advantage of plastic scintillators is their fast response time, making them suitable for high-count rate experiments:

- Rise Time (t_r): Extremely short, typically between 0.5 ns and 0.9 ns.
- Decay Time (τ): Ranges from 1.8 ns (for fast-timing models like BC-404/EJ-204) to 2.1 ns (for general-purpose models like BC-408/EJ-200).
- Pulse Width (FWHM): Approximately 2.2 ns to 2.5 ns.

To ensure maximum light collection and protect the detector from environmental interference, the scintillators undergo a specific wrapping process. The plastic material is first wrapped in aluminum foil that acts as a specular reflector. Then a layer of black vinyl or black tape is applied over the aluminum to provide a light-tight seal. This prevents ambient light from entering the system, which would increase the background noise (dark current) **NON È DARK CURRENT** or saturate the PMT. The scintillator is coupled to the PMT using a BC-802 Tapered Light Guide. Its geometry allows for the transition from the rectangular shape of scintillators to the circular window of the PMT.

The system uses the Hamamatsu R1924A-01 (or similar) as PMT, a 1-inch diameter tube designed for fast timing and prepared for positive high voltage (HV). **CREDO SOLO PER P&G, M?**

SE QUESTE INFO VENGONO DALLA DOCUMENTAZIONE, QUESTO È VERO SOLO PER MINERVA

4.2 Supply voltages & Thresholds

We were now able to test the three scintillators we had: Partenope, Minerva, Giunone. The first interesting thing to study was the relation between number of events, supply voltage and threshold of reading electronics. For a good scintillator-PMT apparatus we expected the output signal to be proportional to the energy released in the interaction by a particle. Thus, at fixed supply voltage, we would expect the higher the threshold is after the PMT, the fewer events measured by a scintillator. For low threshold we expected many counts due to natural radioactivity, but we were interested in measuring the decay of the muon, and for this to happen they must have energy of over ~ 2 GeV **CONTROLLO VALORE (DOVREBBE ESSERE ATTORNO AI 500MEV IL MASSIMO DELLA RADIOATTIVITÀ)**. This energy is high enough to exclude natural radioactivity, so we expected some kind of plateaux in the region of interest (no more natural radioactivity decays, constant rate of muons decays).

We first made some preliminary measurements. These measures were taken varying both the supply voltage and the threshold of the discriminator at the same time (or only one of them) for the three scintillators. These data helped us understanding the number of events we would expect and how much it should have been varying to different initial condition. The first idea was to map the two dimensional space of tension and threshold at the same time, but we soon realized it was a waste of energies and time. Thus, we decided to test the distribution of events at fixed supply voltage increasing the threshold. We chose the supply voltages by visually analyzing the output of PMTs at the oscilloscope and maximizing it with respect to the maximum input the discriminator accepted (250 mV). Then, eventually, we would have lowered the supply tension and retake the measures. In this way, we were sure to respect the requirements of the instrumentation and be able to test all the scale of threshold our system had, searching for the plateaux. This was done for each scintillator.

For what it concerns Giunone and Partenope scintillators, from Figure 17 it is evident that in this setup there were no plateaux. We made some hypotheses: it could be that we missed the plateaux because we took too big steps in threshold resolution or that it was at lower or higher threshold we couldn't set or it could be that the supply voltage wasn't enough high. We also studied the coincidences between the two trying to maximize

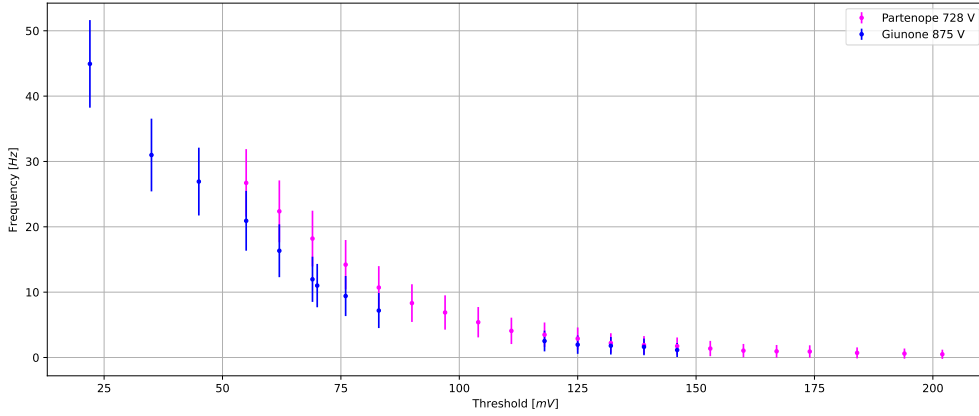


Figure 17: Partenope and Giunone's count vs threshold

the ratio of coincidences and total number of events. But after some measurements we realized this procedure was too messy and wasteful because it depends on the thresholds and supply voltage of both scintillators. Plus, it also could lead us to some misinterpretation of the data because we only measured the double coincidences, so they were affected by casual coincidences and natural radioactivity (since we did not know the number of events from muons' interactions). After having denied the idea of having missed the plateaux because of far too big steps, we thought that maybe the supply voltages of our instruments weren't enough, so PMTs couldn't pre-amplify correctly the signal, which could lead to inaccurate signals. Thus, we wanted to test our detectors at higher supply voltages, but this couldn't be done with our setup. So we had to change our discriminator with another one of different model (from CAEN Mod. N 417 to CAEN Mod. N 96) . With these new instrument we could increase the supply voltage for Partenope and Giunone.

We were now able scan the number of events respect to the thresholds at two different voltages. After analyzing the output of the detectors at the oscilloscope, we noticed that in order to have similar responses to the same input Giunone detector had to be powered with ~ 100 V more than Partenope. We decided to have similar impulses to the same particle interaction.

As it can be seen from the plots (Figure 19 and Figure 18), for each of the two scintillators, we scanned the threshold at two different voltages: for the higher ones tested, we were able to recognize a plateaux for threshold between 130 mV to 175 mV. The plateaux isn't perfectly flat, but we can expect to cut some events (due to the

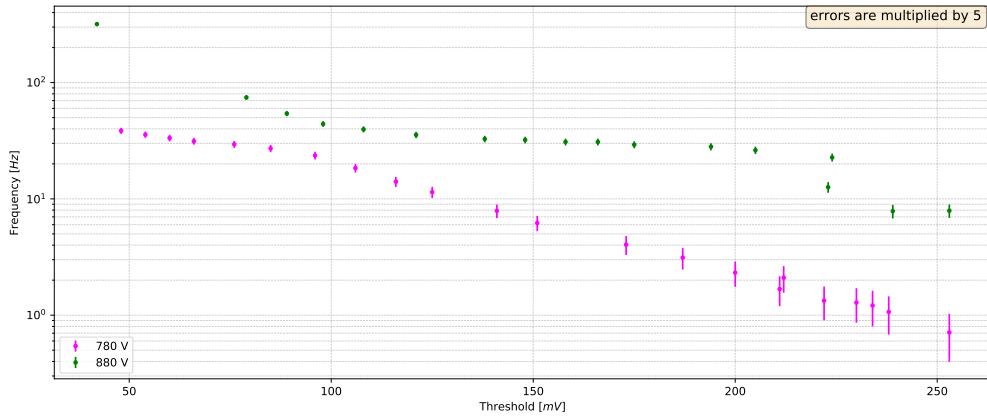


Figure 18: Partenope's threshold

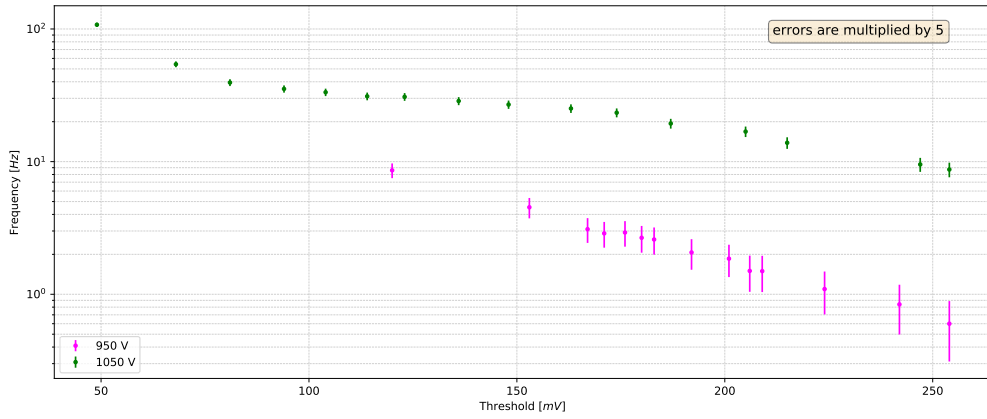


Figure 19: Giunone's threshold

energy distribution or fluctuations) going at higher thresholds. Anyway, we decided to set the threshold at half of the plateaux.

Minerva detector was already at the max supply voltage we could set (for PMT requirements), so we didn't change its one. As can be seen from the plot (Figure 20), for this scintillator we found some kind of plateaux. The plateaux was observed for Minerva scintillator from 157 to 186 mV. We were quite surprised to have found it so easily with this scintillator because we notice quite a few fluctuations in its response (afterpulses and overshooting).

TODO: add ref to Minerva fluctuation

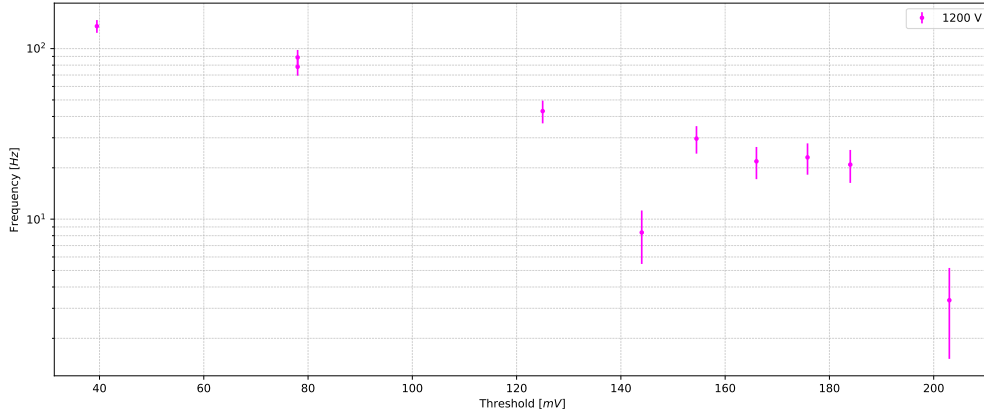


Figure 20: Minerva's count vs threshold

4.3 Efficiency

Having decided the supply voltages and thresholds of the setup, we then decided to study the efficiencies of our instruments: we wanted to test our scintillators to understand the fraction of event registered respect to the number of total interactions. We did this for different configurations to verify also that no scintillator had any “dead zones”. We couldn't know the absolute number of interactions (it is a probabilistic number), but we estimate it by using the counts of coincidences of the upper and lower scintillators. We defined

$$\varepsilon = \frac{\# \text{ triple coincidences}}{\# \text{ upper and lower scintillators coincidences}} \quad (4.1)$$

To do so, we put in the middle the scintillator we wanted to test and above and below the others. For a fixed time, we measured the number of triple coincidences of the three instruments and the number of double coincidences between the upper scintillator and the lower one. We then moved at fixed steps the central scintillator and retake the measurement. We ran a simulation to understand the number of data lost for solid angle (considering different distances among them) and we compared the results. Eventually, we repeated the procedure with the other two scintillators.

4.3.1 Simulations

We ran two simulations: the first was a Python simulation in which we considered only the solid angle factor, the other was a Geant4 simulation in which we simulate real interactions. For the first one, we wanted to have an order of magnitude of muons that

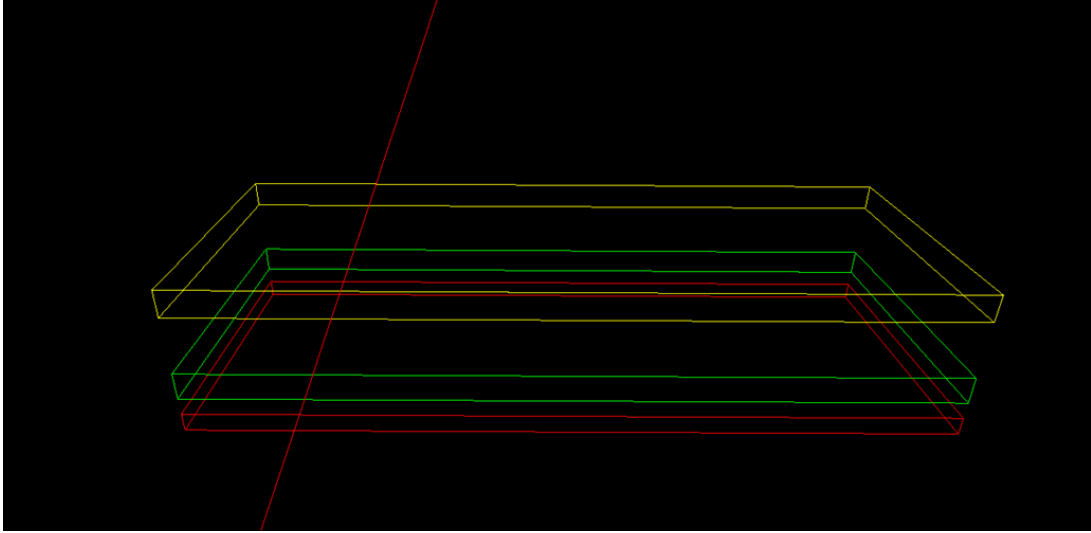


Figure 21: Example of Geant4 simulation run: Minerva (yellow), Giunone (green), Partenope (red)

interact with the upper and lower scintillators while misaligning the three instruments. With this, we did not consider a minimum path and energy released in each of them, but simply if a muon crossed the surface of the scintillator. The second one was a more complete simulation, where we could set energies of particles, kind of interactions, minimum energy released. The two simulations aren't fully compatible, but we do trust more the Geant4 simulation because of its reliability and completeness. With this last simulation we also set different muons distributions (as described in Sec. 1.3)) to understand which one cosmic muons have.

We measured relative distances of scintillators along vertical axes in all the configurations we used for this part of experiment and we implemented this feature in each simulation.

4.3.2 Partenope & Giunone

For this kind of scintillators we did two different simulations in which we varied the length of this detector: in the first simulation it was 80 cm, in the second one it was 75 cm. Since we do not have any technical scheme of these detectors, we tried to understand how much 5 cm of light collector near the PMT end of the scintillator would affect the measure. As can be seen in the following plots (Figure 22 and Figure 23) the difference is very small, almost imperceptible. In these simulations we tried also different angular distribution of cosmic muons respect to the azimuthal angle, but none of them is really compatible with what we found experimentally.

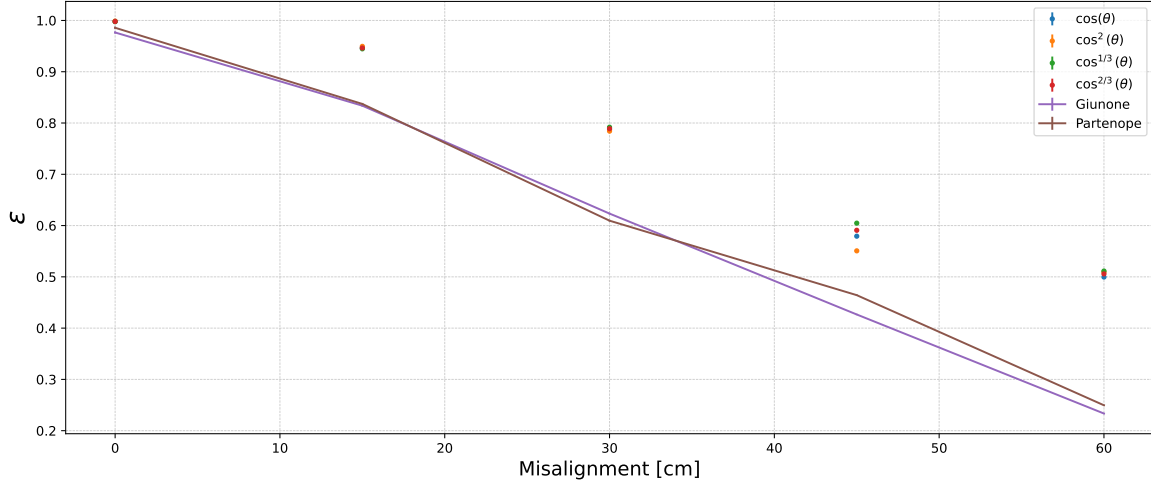


Figure 22: Partenope and Giunone resolutions, 80 cm length

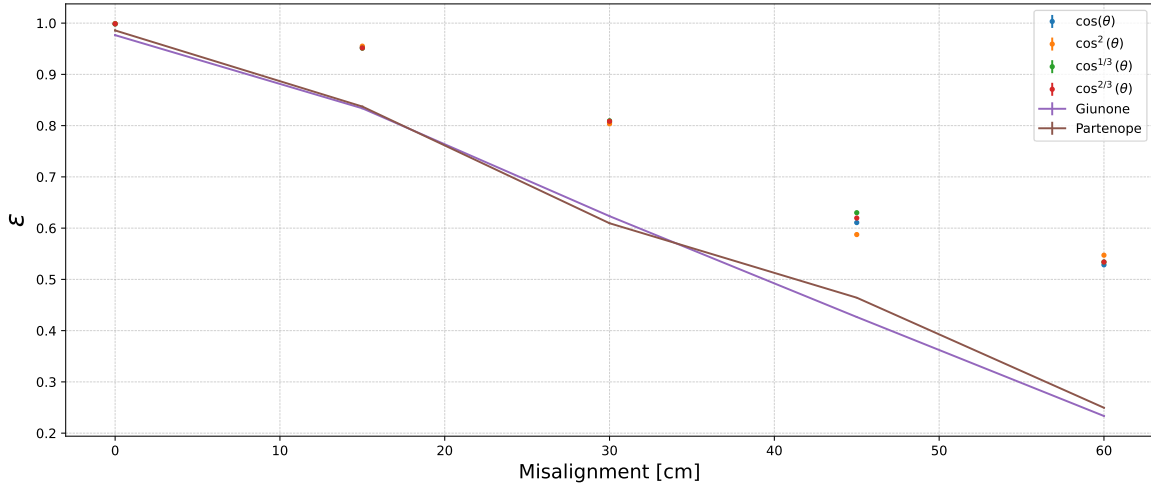


Figure 23: Partenope and Giunone resolutions, 75 cm length

If we define the geometrical efficiency as the fraction $\frac{\varepsilon_{\text{measured}}}{\varepsilon_{\text{theory}}}$, with ε defined in Eq. (4.1), both Partenope and Giunone's plots have an unexpected shape (see figures below). The efficiency drops almost linearly with the misalignment of the middle scintillator, even if it is about 98% (Giunone) and 99% (Partenope) when aligned. This makes us think that there can be some dead zones in this two scintillators, but we can't locate them inside the detector. This dead zones can be either non-scintillating part of the detector or we misinterpreted the real size of the scintillator respect to the measure of the scintillator with light guides and wrapping material around it.

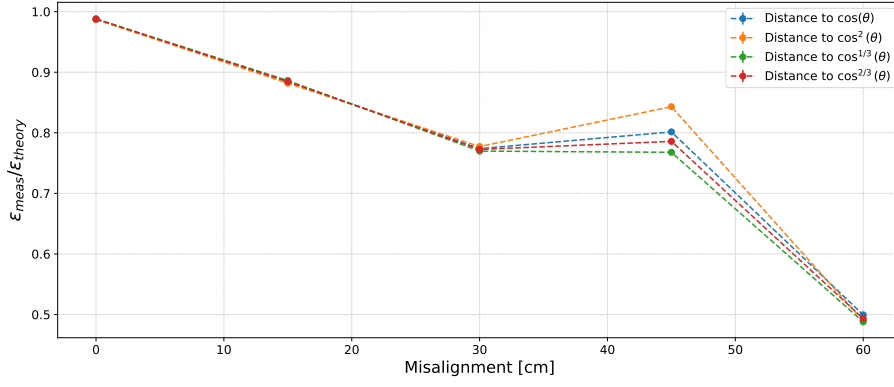


Figure 24: $\varepsilon_{\text{meas}} / \varepsilon_{\text{theory}}$ for Partenope, 80 cm length

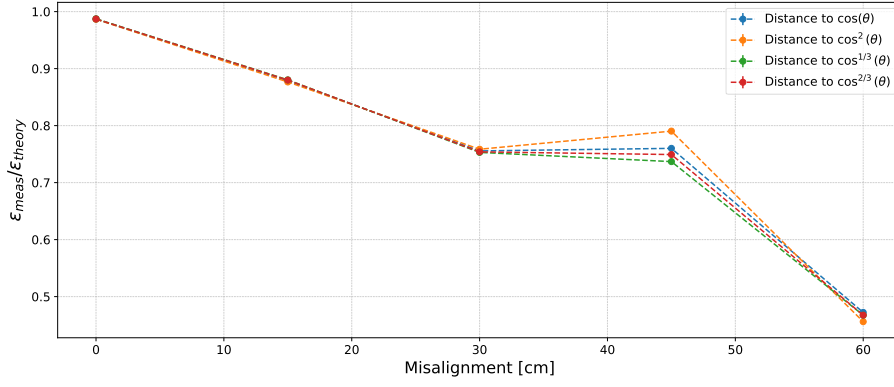


Figure 25: $\varepsilon_{\text{meas}} / \varepsilon_{\text{theory}}$ for Partenope, 75 cm length

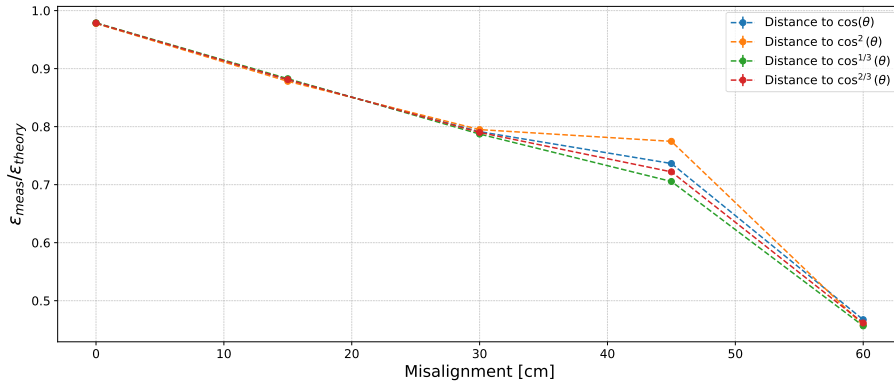


Figure 26: $\varepsilon_{\text{meas}} / \varepsilon_{\text{theory}}$ for Giunone, 80 cm length

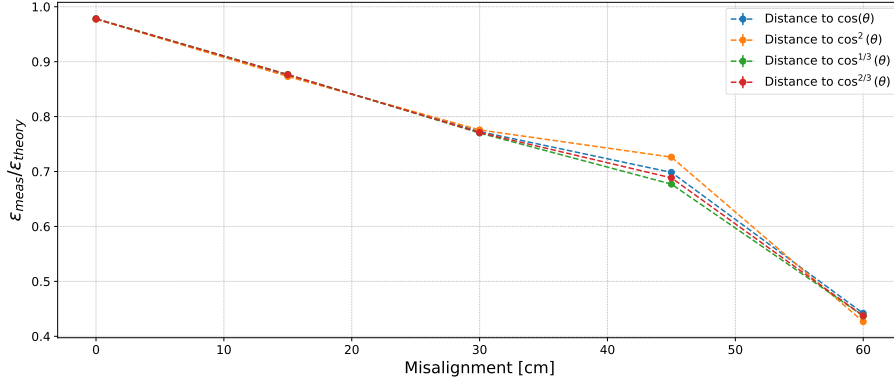


Figure 27: $\varepsilon_{\text{meas}} / \varepsilon_{\text{theory}}$ for Giunone, 75 cm length

Even though there are this dead zones in this two scintillators, we are satisfied by the high efficiency when they are aligned because this is the setup for then next measures.

4.3.3 Minerva

As we can observe also the plots for Minerva scintillator (Figure 28 and Figure 29): this detector had a resolution of about 89% when completely aligned with the others. The “75 cm” label indicates that the length of Partenope and Giunone is 75 cm for that simulation because of what we said in the previous paragraph. The resolution, then, remains constant or slightly increase until 45 cm of misalignment and then it drops at about 60-70%.

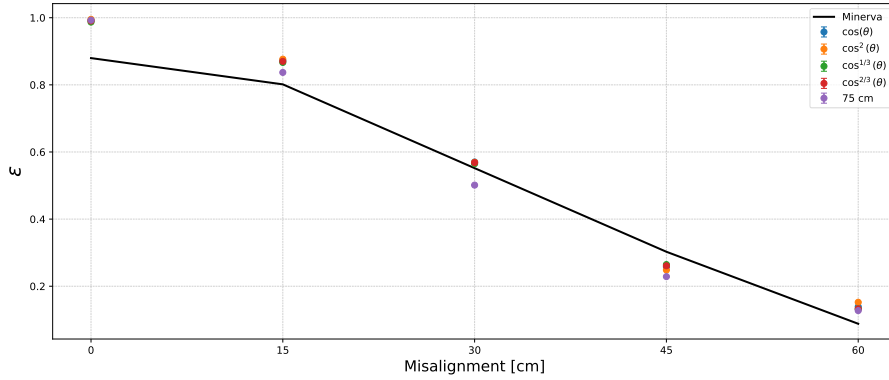


Figure 28: Minerva resolution

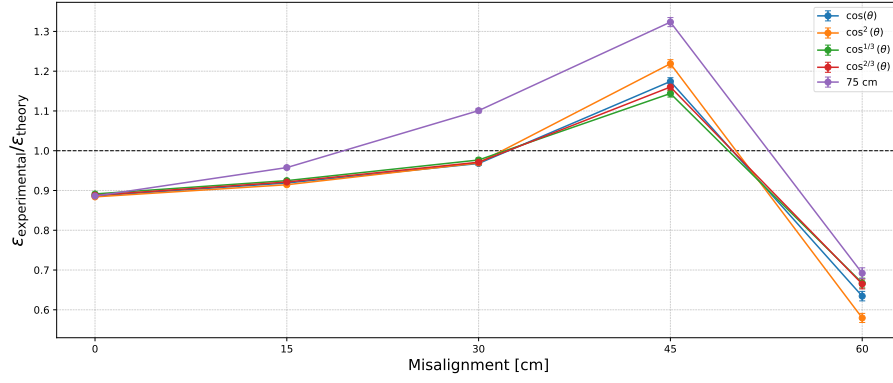


Figure 29: $\varepsilon_{\text{meas}} / \varepsilon_{\text{theory}}$ for Minerva

There is a sort of bump in this last plot where the efficiency is greater than 1, which means we measured an higher count of triple coincidences than the one we would expect due to solid angles. We can see a similar pattern also in the same plots for the other two scintillator. We can address this to the problems we saw with Minerva's PMT or it might be some interference of the PMT while it is in the middle of the other scintillators. Also here we see that the dimensions of the scintillators are important, so ignoring the real dimension of two scintillators doesn't help understanding how performative Minerva is.

5 Data Analysis

The data analysis has been made using iMinuit, we used poissonian errors for each bin and we minimised the negative log likelihood.

5.1 Initial issues

We found three issues with this method:

- what to use to quantify the goodness of the fit
- how many bins would each histogram need
- the model to choose

5.1.1 Chi square as estimator of the goodness of the fit

Although data inside each bins are distributed as a Poisson distribution, with enough data we can fall back under the hypothesis of data distributed as a normal distribution. If we assume this hypothesis as valid, what we found is a minimum number of events per bin to approximate the poissonian distribution to a normal distribution and, if we respect this constraint, we can say that $\tilde{\chi}^2$ is a good estimator of the goodness of the fit similarly to what happens when using Least Squares method.

5.1.2 Model of the interpolation

We were uncertain about wether to consider an uniform noise background or to interpolate without it. By letting the frequency between exponential distributed points and uniform distributed points, we found that the noise frequency is around 5%. Although it might be small, it has a relevant impact in the error estimation of the variable: if we consider an uniform background, both relative errors and $\tilde{\chi}^2$ decrease, therefore we decided to keep it.

FOTO CON GLI ERRORI DELLA TAU NEI 2 MODELLI NEI VARI DATASET

5.1.3 Bins choice

To choose how many bins to give each histogram, we used $\tilde{\chi}^2$ to see how the interpolation behaved at different bins width. We found it generally to be between 60 and 100 bins but the specific number changes from dataset to dataset.

5.2 Vacuum measurements

As said in [CHAPTER 1 REF](#), in the presence of only the scintillator, the variation of the muon's decay is very slightly different from the one in the vacuum. This difference cannot be seen in our data sample since it is dominated by statistical uncertainties. We therefore compare the result of the fit to the decay time of the muon in vacuum. [REPHRASE](#)

As fit function we use a combination of exponential and a uniform background, normalised on the dataset, as follows:

$$f_{\{\text{vacuum}\}}(t; f_{\text{exp}}, \tau) = N \left[f_{\text{exp}} \cdot p_{\text{exp}}(t, \tau) + (1 - f_{\text{exp}}) p_{\text{uniform}} \right] \quad (5.1)$$

Due to the presence of unexpected signals at the start of the trace, and due to edge effect at the end of the dataset, we decided to apply two cuts to the dataset. To choose the optimal position where those cuts have to be performed, we ran some interpolations varying the time interval in which we consider the data. From every fit, we took the $\tilde{\chi}^2$ and showed it in Figure 30.

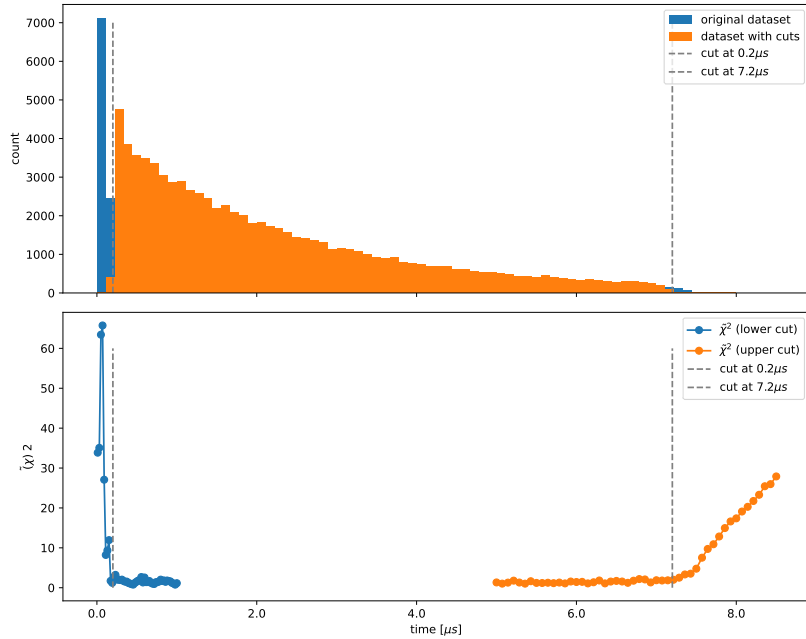


Figure 30: On top: Raw data collected and the data taken after the cuts.

On bottom: $\tilde{\chi}^2$ for different cuts at the start of the dataset (blue), and at the end of the dataset (orange). We can see how at the start and at the end of the dataset there are data that notably worsen the interpolation. We believe the signals in the very first $0.2\mu\text{s}$ are due mainly to Minerva's PMT fluctuations superimposed on muon's decay signals. The noise over

$7.5\mu s$ is due to a small jitter in the end gate of the timer that therefore decrease the number of data acquired in that region and therefore decide to remove them.

In order to choose the number of bins N to use, we calculated the $\tilde{\chi}^2$ of the dataset for different number of bins and chose the one that minimises it.

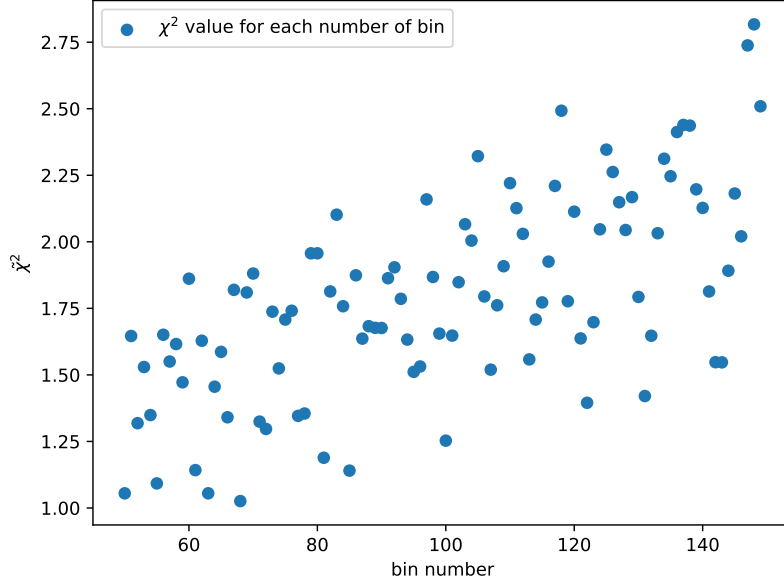


Figure 31: $\tilde{\chi}^2$ for different number of bins.

Since we don't see a clear pattern varying the The number of bins taken is 68, for the range $[0.2, 7.5]\mu s$, causing every bin to have a $10ns$ width. In the Figure 33 we show the final fit of the dataset, together with the percental deviation of the data from the interpolated function. In Table 6 contains the fitted parameters of the free decay, with their errors.

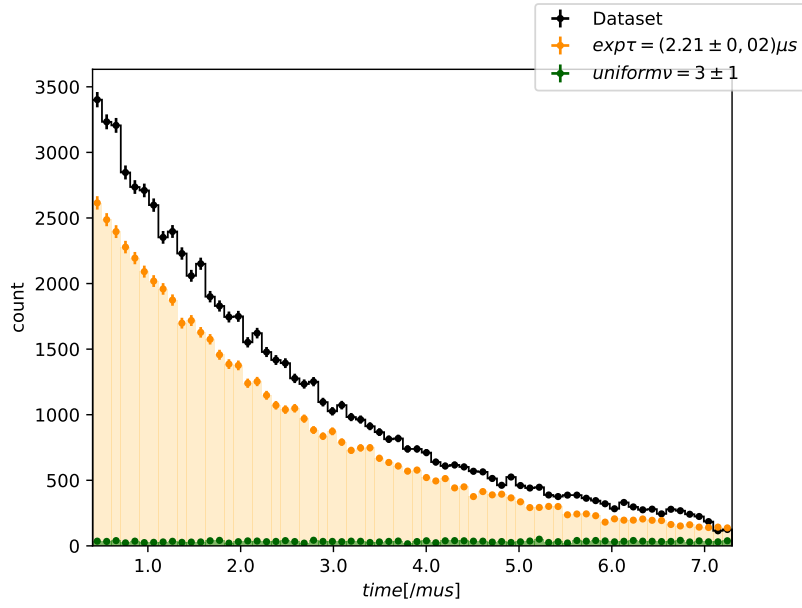


Figure 32: interpolation divided in components. The data of each component is simulated with the data extracted from the interpolation

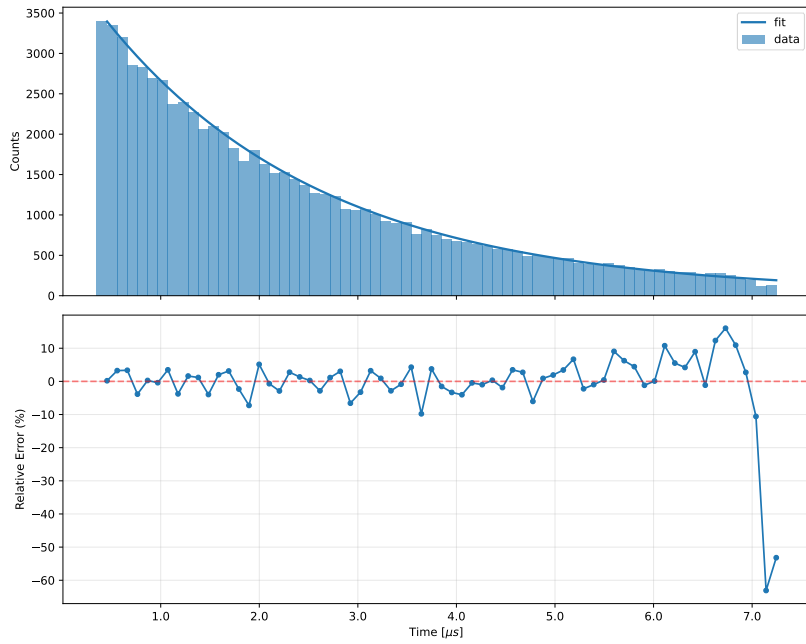


Figure 33: On top: The fitted dataset, and the resulting interpolation function.
On bottom: Relative deviation of the data respect to the fit.

Parameter	Value	Uncertainty
f_{exp}	0.967	0.010
τ	$2.22 \mu s$	$0.03 \mu s$

Table 6: Vacuum fit parameters

From the values in Table 6, and using the function in Eq. (5.1), we can derive the amplitude A of the uniform background, and its error, obtaining $A = 1 - f_{\text{exp}} = 0.033 \pm$

0.010. Although the uniform background noise is low, it is not compatible with zero, therefore is not negligible.

TODO: test di ipotesi?

In Figure 34 we also show the Negative-Log-likelihood of the parameter τ , and the comparison with the expected value.

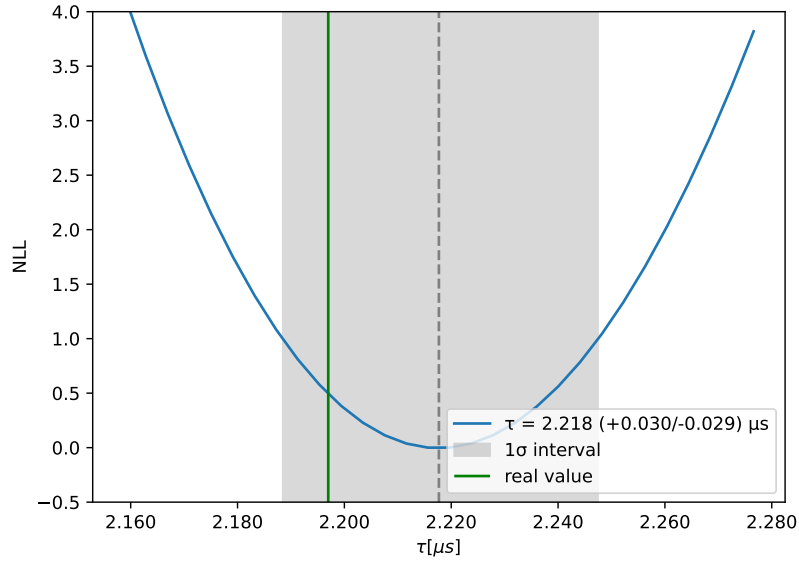


Figure 34: Negative-LogLikelihood of the parameter τ

As we can see, the expected value is below 1σ distant from our measure, therefore is compatible.

TODO: test di ipotesi?

As a further check for our setup, we looked at the up-down asymmetry of this dataset, defined as in Eq. (1.12).

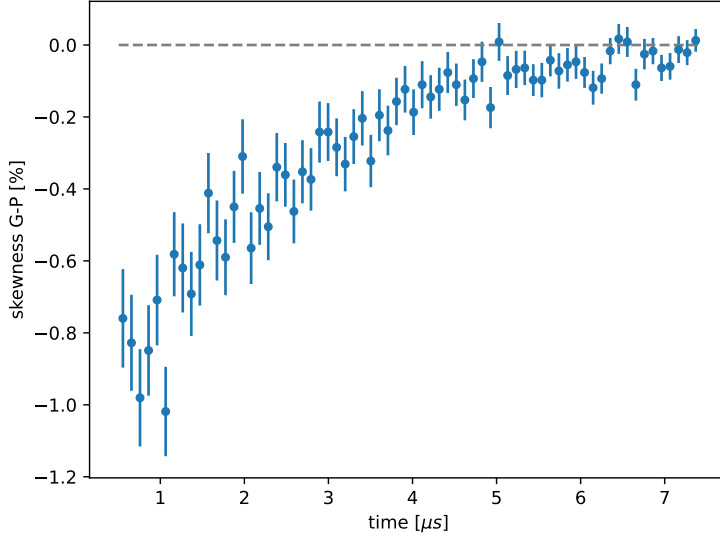


Figure 35: Up-Down asymmetry of vacuum dataset

We can see in Figure 35 how the asymmetry is in favour of the lower scintillator, although is consistently inside the 1% range from zero. Therefore we can consider it negligible.

5.3 Salt measurements

We changed our setup to accommodate some salt in between Minerva and Partenope. This has brought a different behaviour to be studied: the decay of muons in salt due to the interaction with the nuclei. This interaction is way more frequent than in carbon and it creates a greater variation with respect to the mean lifetime of a muon.

To study this behaviour, we modified the model used previously in Eq. (5.1). In particular, we switched the exponential part with a double exponential determined by another two parameters of the interpolation

$$p_{\text{exp}} = f_{\text{relative}} p_{\text{exp}}(t, \tau_{\text{void}}) + (1 - f_{\text{relative}}) p_{\text{exp}}(t, \tau_{\text{NaCl}}) \quad (5.2)$$

Similarly to what we did with vacuum measurements, we used $\tilde{\chi}^2$ to decide which bin width and how much to cut at the start and at the end of the dataset. Since the setup is the same as before, here we expect to apply cuts in points similar to the ones in Figure 30.

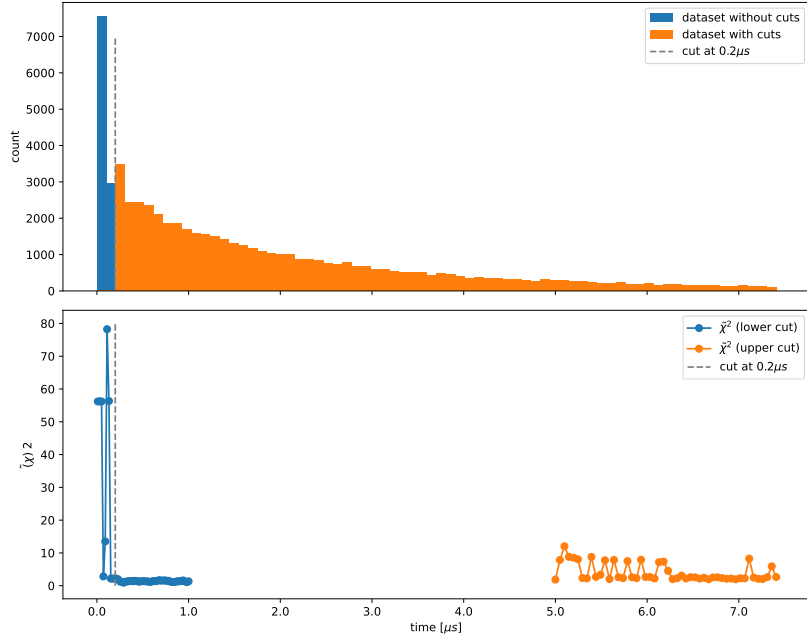


Figure 36: On top: Raw data collected and the data taken after the cuts.

On bottom: $\tilde{\chi}^2$ for different cuts at the start of the dataset (blue), and at the end of the dataset (orange).

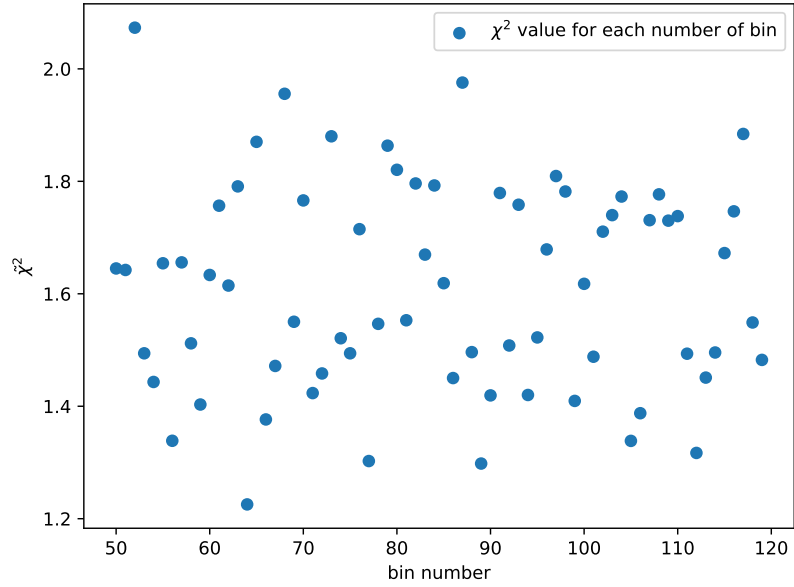


Figure 37: $\tilde{\chi}^2$ for different number of bins.

Following the same logic as the previous case, we opted for a bin width slightly larger than in the previous case with a bin number of 64 and a bin width of $11.4ns$ in the range $[0.2, 8]\mu s$.

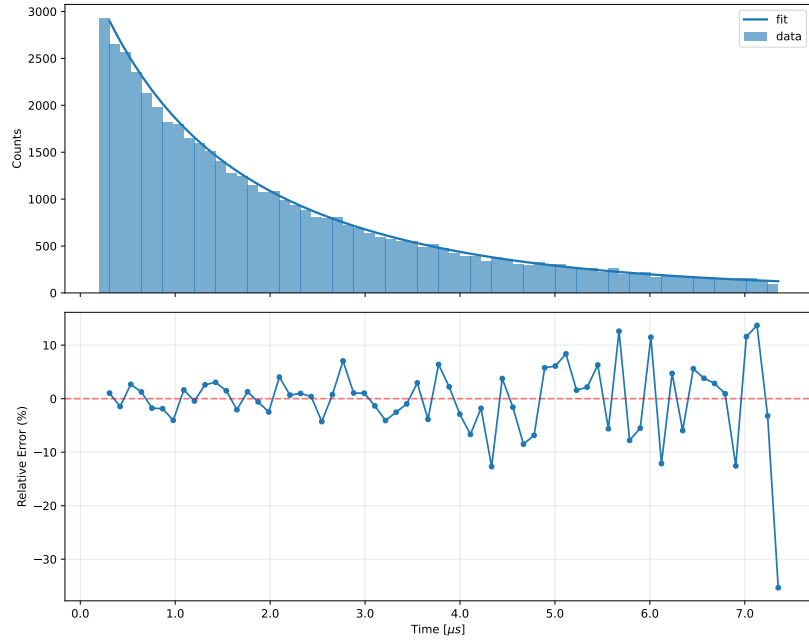


Figure 38: On top: The fitted dataset, and the resulting interpolation function .
On bottom: Relative deviation of the data respect to the fit.

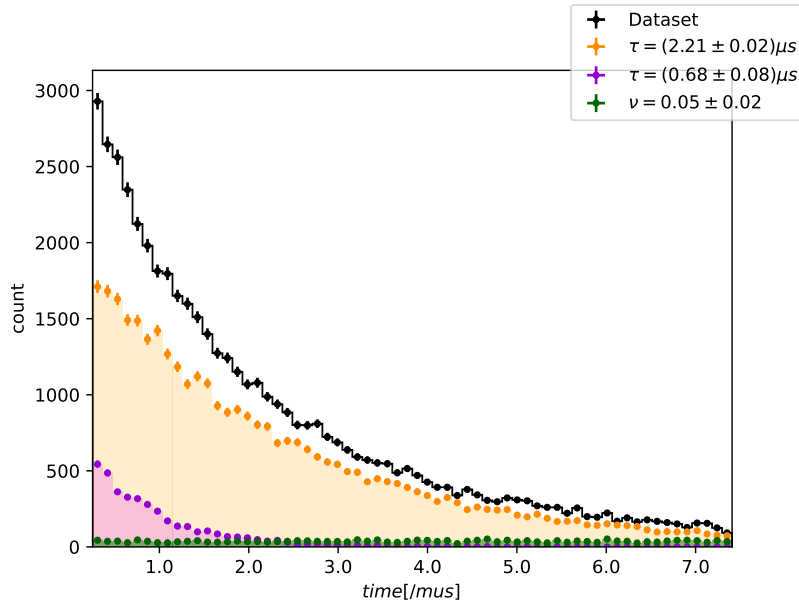


Figure 39: interpolation divided in components. The data of each component is simulated with the data extracted from the interpolation

As we can see, here clearly a second exponential is present in addition to the noise background that we expect to be about constant in all the measures taken.

Parameter	Value	Uncertainty
$f_{\text{exp } 1}$	0.8403	0.0229
$f_{\text{exp } 2}$	0.1093	0.0165
τ_1	$2.2177\mu s$	$0.022177\mu s$
τ_2	$0.71840\mu s$	$0.084299\mu s$

Figure 7: Salt Fit parameters

In the interpolation we opted for fixing the components in common between the two measurements. Them being the uniform background noise and the characteristic of the exponential for the decay of the muon with no interaction. With this parameter fixed, we measure a mean lifetime of the muon in the salt compatible with the expected value that we assume known without error.

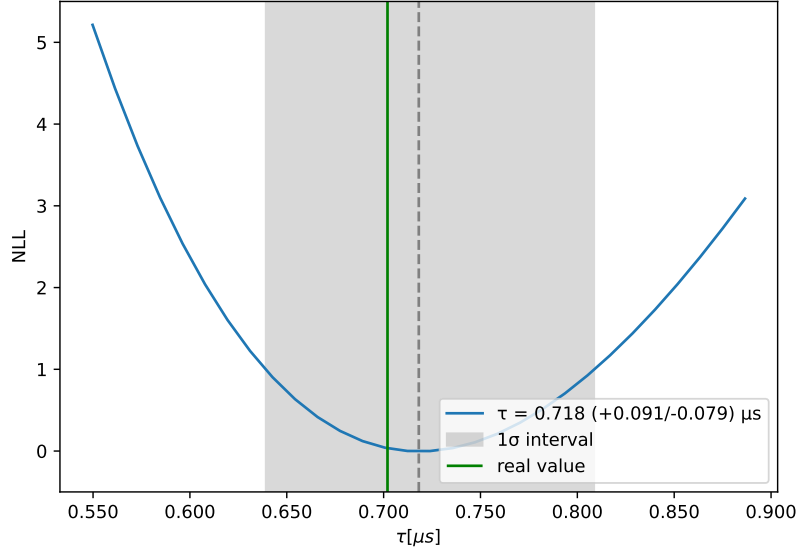


Figure 40: Negative-LogLikelihood of the parameter τ_2

Moreover we kept checking whether the addition of material inside the setup broke the symmetry we found in the previous part.

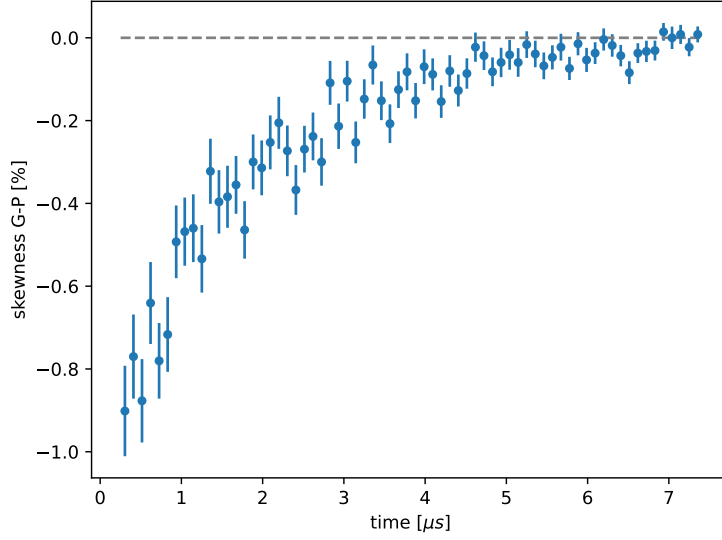


Figure 41: Up-Down asymmetry of salt dataset

What we found is an asymmetry comparable to the one before.

5.4 Aluminium measurements

Lastly we substitute salt with slabs of aluminium. This change, as said in chapt [REF](#), should return data close to the ones gathered in salt but with different frequency and lifetime since aluminium has a higher atomic number.

In this measurements we used the same interpolation as in salt in Eq. (5.2)

Analogously to what has been done previously, we used $\tilde{\chi}^2$ to decide bin number (or bin size) and which cuts to apply.

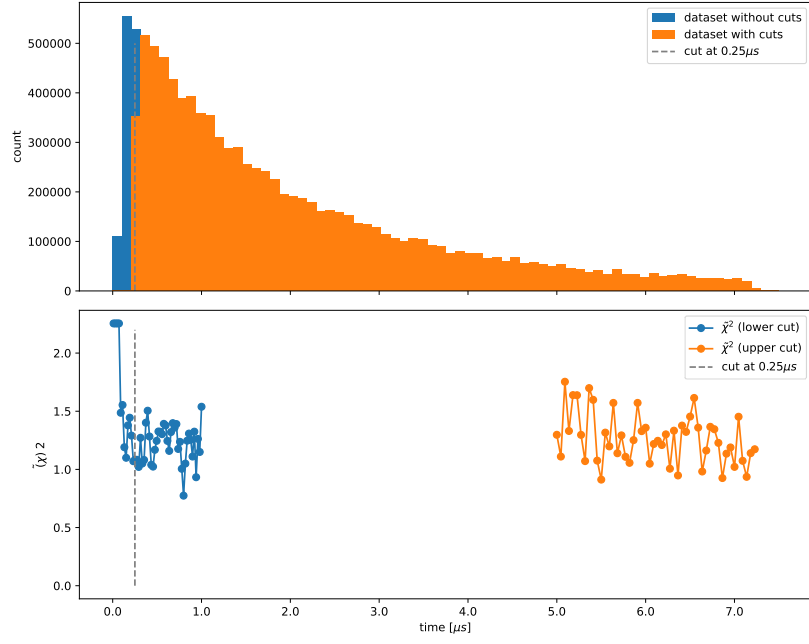


Figure 42: On top: Raw data collected and the data taken after the cuts.

On bottom: $\tilde{\chi}^2$ for different cuts at the start of the dataset (blue), and at the end of the dataset (orange).

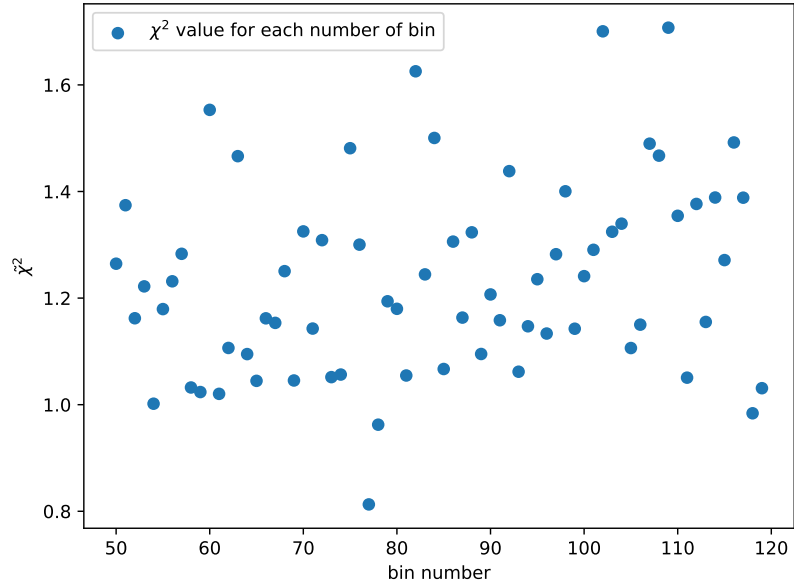


Figure 43: $\tilde{\chi}^2$ for different number of bins.

This time, the shape of the dataset has notably changed and with that also the bin number. We chose 94 bins for the range $[0.25, 8]\mu s$ for a bin width of $82ns$

In this case we delayed slightly the electron signal and this gave a huge improvement in the cutting zones. While in the bin width the range is close to the others with 78 bins.

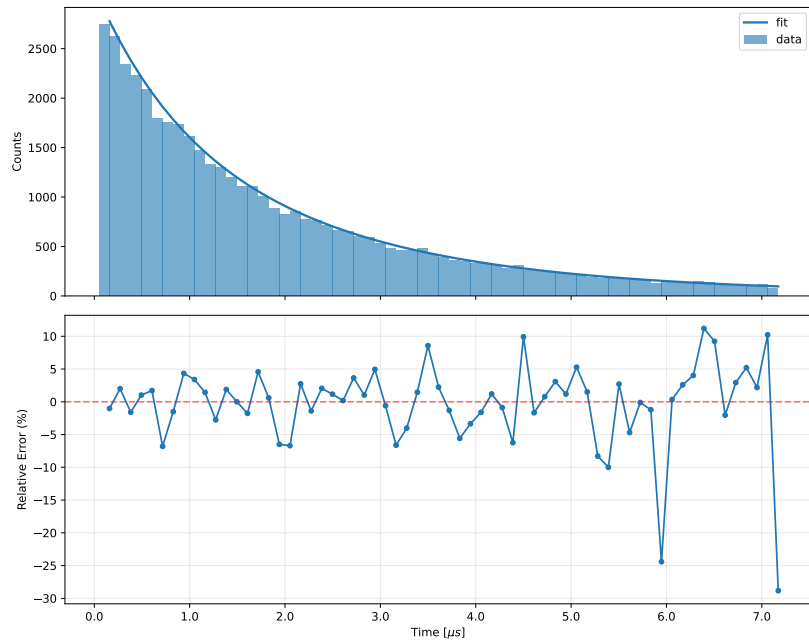


Figure 44: On top: The fitted dataset, and the resulting interpolation function **REF?**.

On bottom: Relative deviation of the data respect to the fit.

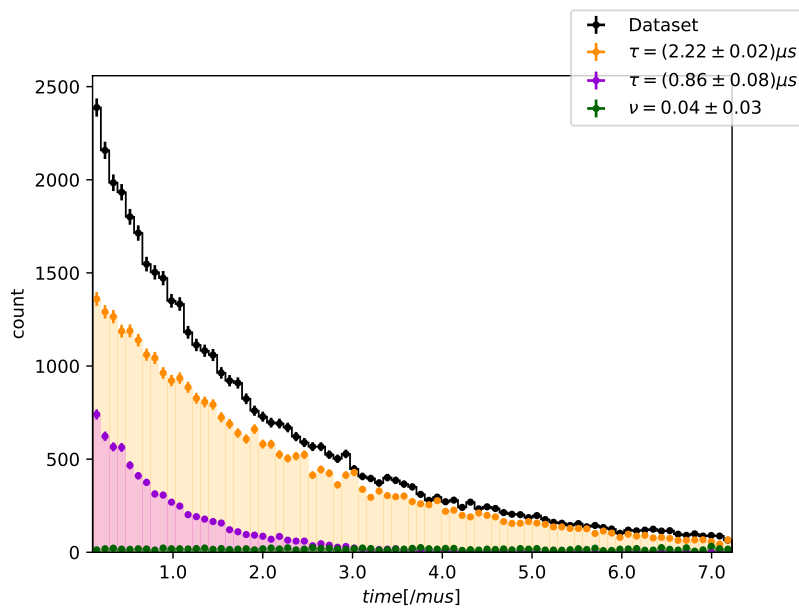


Figure 45: interpolation divided in components. The data of each component is simulated with the data extracted from the interpolation

As previously, a second exponential is present and the uniform distribution is around the same percentage

Parameter	Value	Uncertainty
$f_{\text{exp } 1}$	0.7903	0.0324
$f_{\text{exp } 2}$	0.1790	0.0249
τ_1	$2.22\mu s$	$0.02\mu s$
τ_2	$850ns$	$80ns$

Figure 8: Fit Parameters

Similarly to what done previously, we fixed the common factors between interpolation. Doing so we found a value of the lifetime of the muon when interact with the nuclei of aluminium that is compatible to what has been found in chapter [REF](#)

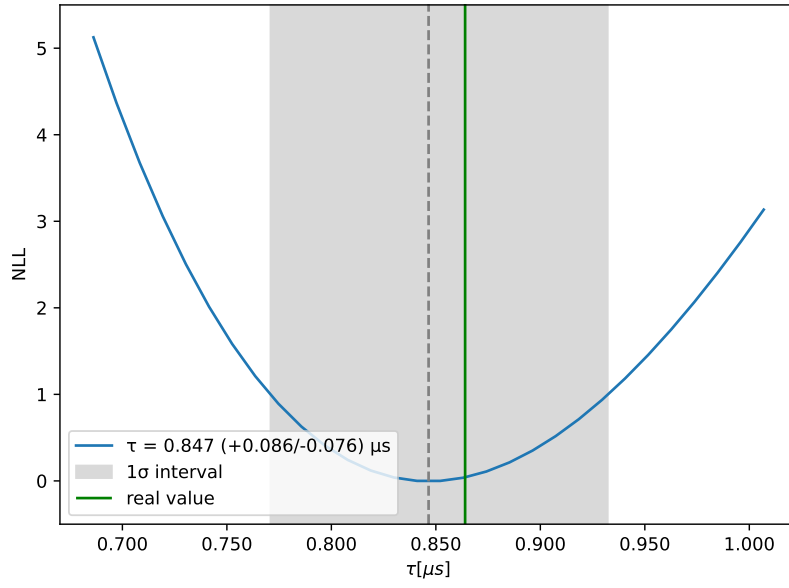


Figure 46: Negative-LogLikelihood of the parameter τ_2

The changes in the setup improve further the symmetry of the setup. In fact, in this dataset we are close to the 0.5% for the

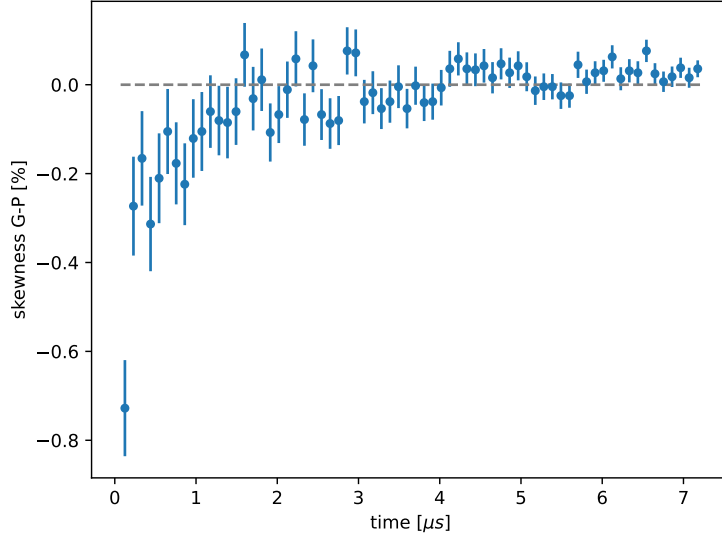


Figure 47: Up-Down asymmetry of aluminum dataset

5.5 Parity Violation

As discussed in section Sec. 1.7, in order to see the effects of parity violation with our setup we have to insert the middle scintillator (the one in which happens the decay) in a magnetic field such that the spin of the decaying particle precedes around the direction of the field. Because of mechanical and schedule issues we had to join another same experiment and to study parity violation with a different setup: from our previous experiments we only used Partenope scintillator. All the characterisation and tuning of the new setup was done by the other group **AGGIUNGO REF O NOMI DEGLI ALTRI**.

Before starting to take measurements, we need to study the magnetic field generator and be sure that the magnetic field is constant and stable under Biot-Savar law.

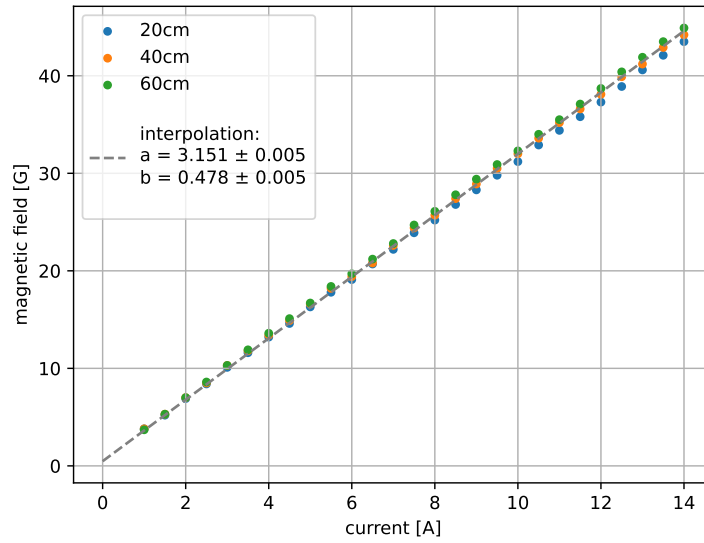


Figure 48: Intensity of field in 3 points of the coil given different currents

Before the beginning of the measures of the parity violation, we tested the uniformity and the stability of the magnetic field inside the magnetic field generator (it's just a coil enough big to contain the scintillator) with a magnetometer. Results are in Figure 49.

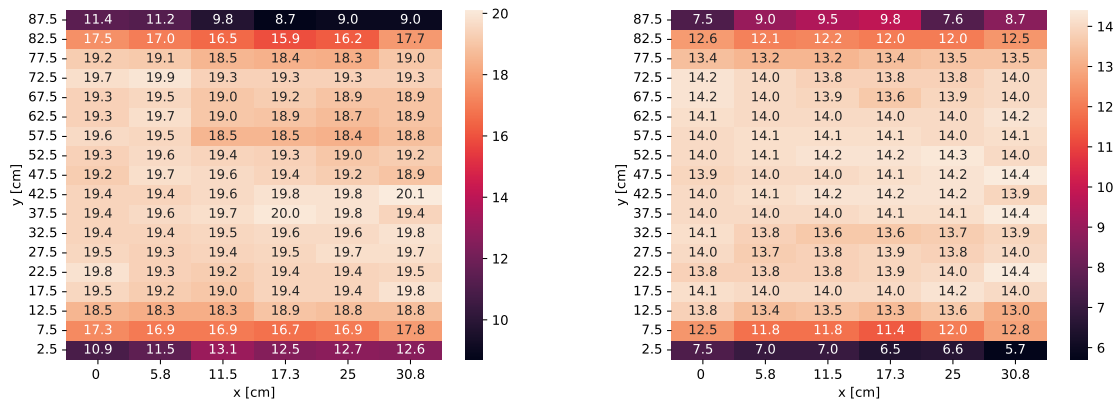


Figure 49: Uniformity test results of magnetic field. The scale is in Gauss. All data are $\pm 0.2G$,
left figure: 6A , right figure 4.5A

Multiple trials has been made but we finally chose to take measurements at a current of 4.5A

TODO: ricordarsi foto precessione del muone

5.6 Measurements with no magnetic field

Since neither of the two group has ever used this setup, we opted for a preliminary measure with the coil but without turning the magnetic field on. This will give us a confirmation of the behaviour of the setup and an initial measurement of the asymmetry of the setup that will need to be detracted from the real measurement.

We proceeded therefore in similar way to what has been done previously. Since the objective of the measurement is not the detection of the behaviour we presumed the decay of the muon in the coil to be negligible and interpolated the data with the model in Eq. (5.1).

We therefore proceeded to decide where to cut the dataset and the optimal bin width.

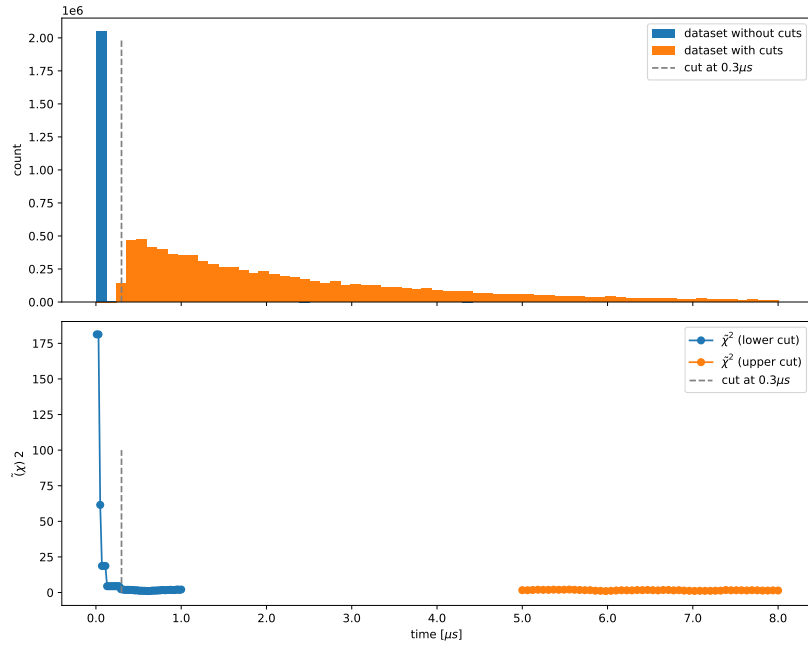


Figure 50: On top: Raw data collected and the data taken after the cuts.

On bottom: $\tilde{\chi}^2$ for different cuts at the start of the dataset (blue), and at the end of the dataset (orange).

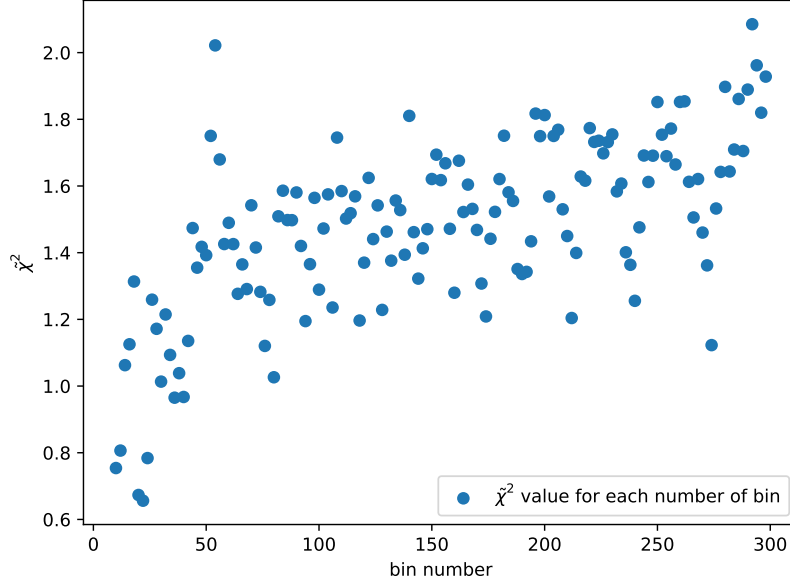


Figure 51: $\tilde{\chi}^2$ for different number of bins.

We see that the optimal number of bins is at 80 between the range $[0.3, 8]\mu s$ with a bin width of $0.1\mu s$.

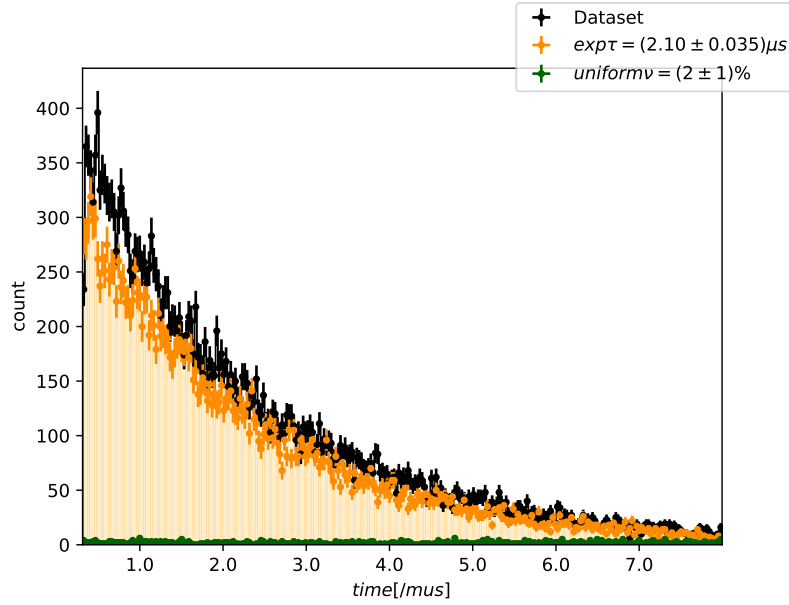


Figure 52: interpolation divided in components. The data of each component is simulated with the data extracted from the interpolation

We see how the uniform is less relevant than the other cases. This gives us reason to think that the uniform noise is mainly given by the presence of Minerva (the thin scintillator). **WE CAN ONLY SAY THAT PROBABLY IT ISN'T PARTENOPE**

Parameter	Value	Uncertainty
freq_exp1	0.9818	0.0106
tau1	2.1024e-06	3.5122e-08

Figure 9: Fit Parameters

In this case we are unable to fix any factors since the setup is different.

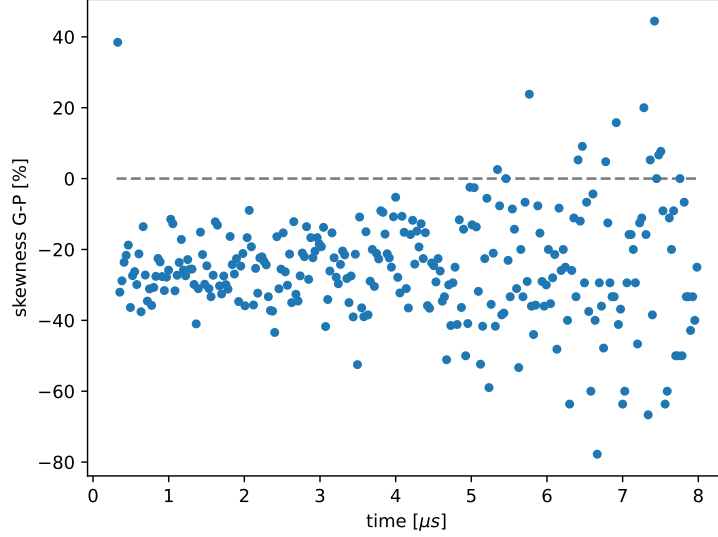


Figure 53: Up-Down asymmetry of the dataset

Finally, we see how in the setup an initial asymmetry towards the bottom scintillator is present and therefore needs to be subtracted from the future dataset.

5.7 Measurement with magnetic field

Turning on the magnetic field, we want to measure the variation of symmetry between the up and down scintillators. If we chose a bin number similar to the one used above, we wouldn't be able to see anything since the difference is negligible with bins of that size. Therefore we opted for a number of bin of 6 to be able to see the phenomena.

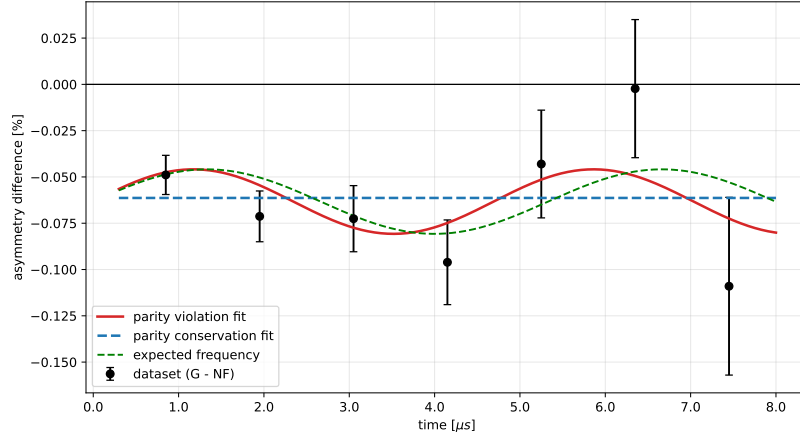


Figure 54: Data asymmetry with both models of parity violation and conservation

As we can see, initially we are not able to determine whether either model is better to describe the dataset. Therefore we use the likelihood-ratio test to see on what probability or dataset can be described by both models.

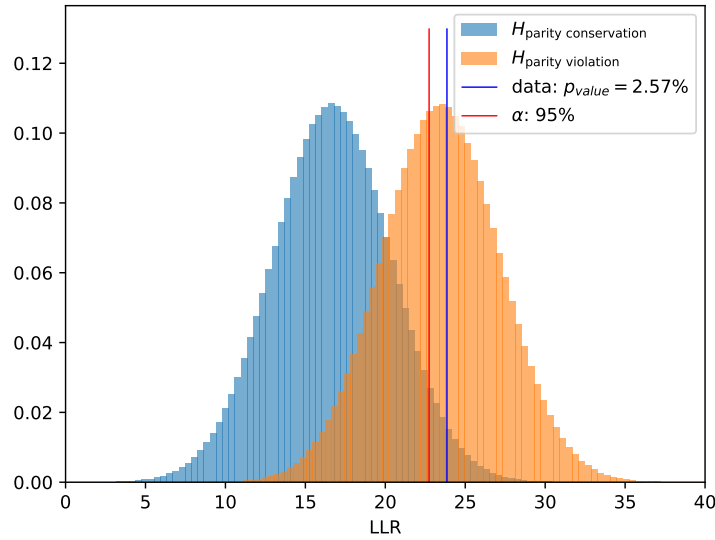


Figure 55: Hypothesis testing using the log-likelihood ratio test,

We simulated data given the known errors of each point and generate the distribution of the likelihood ratio given the different models. What we found is depicted in Figure 55: we see how our model can reject the null hypothesis with 95.20% probability and therefore we can say that we are able to detect parity violation in our data sample.

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